

Report194

Asymptotics for Binary Matrices with Zero Row and Column Slopes

A self contained study of the square diagonal of OEIS A222959

4 October 2026

Abstract

Let C_n count the $n \times n$ binary matrices whose least-squares regression slope is zero in every row and every column. With $S_n = n(n^2 - 1)/12$ for odd n and $S_n = n(n^2 - 1)/3$ for even n , we prove

$$C_n = 2^{n^2} e^{-7/5} \sqrt{\frac{2}{\pi}} \left(\frac{2}{\pi S_n} \right)^{n-1} \left(1 + \frac{171}{350n} + \frac{483051}{245000n^2} + O(n^{-3}) \right).$$

More generally, the factor in parentheses has an asymptotic expansion to every fixed order, with rational coefficients independent of parity, given by a finite Wick pairing and power-sum prescription. The main analytic issue is global localization in a growing-dimensional Fourier integral. We prove a relative exponential bound outside one embedded central tube by a dense arithmetic progression inverse lemma and a defect-core decomposition. The proof includes lattice saturation, Haar normalization, uniform core Gaussian ratios, and uniform Gaussian polynomial moment estimates. We derive parity-aware smooth inverse expansions and rigorous odd-integer threshold envelopes. Exact finite counts and reproducible rational checks are supplied separately from the asymptotic proof. The known small counts are strongly preasymptotic. No practical onset, convergence of the full formal series, or absolute priority is claimed.

[write] **Status and trust boundary, added 6 October 2026, batch 108.** This is bundle Report 194 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a222959-zero-slope-matrices** (provenance, the sources as the write read them, relation to the repository, the collected non-claims and reading conventions in Section 1.1). It is unrefereed and not formalized; its author line is a subtitle and its PDF author field reads “Research report”. *What rests on what.* The proofs of Theorems 1.1 and 1.2 and of Corollary 1.3 are written out in the text and cite no external theorem: the lattice, Haar and Gaussian normalizations, the inverse lemmas, the defect estimate, the dimension-free moment bound and the connected power count are all proved here; the literature of Section 13 is credited as method precedent, not used as input. The value c_1 is computed by hand in Section 9. The value c_2 uses only the leading terms (59) and (60), which the source also obtains by summing the per-graph entries of its table in Section 10 (the write rechecked that sum); the complete polynomials (56)–(57) come from the package’s symbolic computation, replayed at intake. The finite counts are exact computations and are used by no proof. The program behind the source’s “recorded” $n = 9$ count, a C++ meet-in-the-middle enumeration, is not shipped; the intake reproduced every count through $n = 9$ with the shipped Python and with its own program, and, from the OEIS b-file the source could not retrieve, the count $C_{10} = 615667968$ (Remark 12.1). The write also adds Remark 13.1 (the conjecture recorded in A222955 is a theorem of Richomme; a two-line second proof), Remark 11.1 (the inversions of Section 11 against the transseries volume: the smooth inverse is a formal instance, the integer envelopes analogues) and Proposition 14.1 (narrow arrays, and the sharper sector bound $C_n \geq b_n^{\lceil n/2 \rceil}$), with the open questions in Section 14.

Attribution and scope. The matrix sequence is OEIS A222959 [1]. Its one-dimensional row condition belongs to the earlier theory of fair binary words, including Prodinger’s asymptotics and Richomme’s regression interpretation [3, 4]. The Fourier and higher-order Gaussian enumeration strategy is part of the maximum-entropy and Edgeworth framework of Barvinok and Hartigan [7, 8]. The present report supplies a complete problem-specific proof for the simultaneous growing arithmetic-progression row and column constraints. The primary-source comparison in Section 13 is deliberately scoped; a failure to locate an identical theorem is not a certificate of originality.

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1 The counting problem and the main results

All logarithms are natural. All asymptotic assertions refer to $n \rightarrow \infty$; constants may depend on a fixed truncation order, but not on parity. The symbol O_J makes that dependence explicit. No constant below is optimized.

For $n \geq 2$, regress a row $(x_1, \dots, x_n)$ against the positions $1, \dots, n$, including an intercept. Its slope is

$$\frac{\sum_{j=1}^n (j - (n+1)/2)x_j}{\sum_{j=1}^n (j - (n+1)/2)^2}.$$

Thus zero slope is a homogeneous weighted-sum condition. Define the primitive integer weight vector $w = (w_1, \dots, w_n)^T$ by

$$w_j = \begin{cases} j - (n+1)/2, & n \text{ odd}, \\ 2j - n - 1, & n \text{ even}. \end{cases} \quad (1)$$

The entries sum to zero, contain 1, and form a symmetric arithmetic progression of spacing $d_0 = 1$ or 2. Write

$$S = S_n = \sum_j w_j^2 = \begin{cases} n(n^2 - 1)/12, & n \text{ odd}, \\ n(n^2 - 1)/3, & n \text{ even}. \end{cases} \quad (2)$$

Then the square diagonal of A222959 is

$$C_n = \#\{M \in \{0, 1\}^{n \times n} : Mw = M^T w = 0\}.$$

We use $C_1 = 2$, the sequence's convention that its length-one rows and columns are admissible; a regression slope denominator at $n = 1$ is zero. Since $\sum_j w_j = 0$, replacing M by the sign matrix $2M - \mathbf{1}\mathbf{1}^T$ preserves the two nullvector conditions. Consequently $C_n/2^{n^2}$ is the probability that a random sign matrix has this prescribed vector as both a right and a left nullvector.

Define

$$G_n = \sqrt{\frac{2}{\pi}} \left(\frac{2}{\pi S_n} \right)^{n-1} = \left(\frac{2}{\pi} \right)^{n-1/2} S_n^{1-n}. \quad (3)$$

The parity dependence in S_n is essential: removing it would introduce an exponential error.

Theorem 1.1 (All fixed orders). *There is a sequence of rational numbers $(c_j)_{j \geq 0}$, independent of parity, with $c_0 = 1$, $c_1 = 171/350$, and $c_2 = 483051/245000$, such that, for every fixed integer $J \geq 0$,*

$$C_n = 2^{n^2} G_n e^{-7/5} \left(\sum_{j=0}^J \frac{c_j}{n^j} + O_J(n^{-J-1}) \right). \quad (4)$$

Section 8 gives a finite exact coefficient prescription. No convergence of the infinite formal series is asserted. The coefficients c_0 , c_1 , and c_2 are explicitly evaluated in this report.

Theorem 1.2 (Relative global localization). *Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let coefficient variables a, b range over $\mathbb{T}^n$, and put*

$$x_{ij} = w_j a_i + w_i b_j \pmod{1}, \quad F(a, b) = \prod_{i,j} |\cos(\pi x_{ij})|.$$

Let $E = \{aw^T + wb^T : a, b \in \mathbb{R}^n\}$ and $\Lambda = E \cap \mathbb{Z}^{n \times n}$. Let U be the image in E/Λ of $\{X \in E : \max_{ij} |X_{ij}| < 1/16\}$. There is an absolute $c > 0$ such that

$$\int_{\{(a,b): x \notin U\}} F(a, b) da db = O(G_n e^{-cn}). \quad (5)$$

The measure on every coefficient torus is normalized Haar measure.

The second theorem is stronger than locating the exact maximum set of the characteristic function. It controls all approximate modular resonances, relative to the small main mass $G_n = \exp(-3n \log n + O(n))$.

Corollary 1.3 (Parity and the first threshold). *One has*

$$\frac{C_{2m}}{C_{2m-1}} \sim \frac{6e^{-3}}{\pi m^3}. \quad (6)$$

Each parity subsequence is eventually strictly increasing, whereas the combined sequence is not. For all sufficiently large real Y , the first threshold index

$$N(Y) = \min\{n \geq 1 : C_n \geq Y\}$$

is odd. Section 11 gives rigorous upper and lower envelopes for $N(Y)$, including possible near-integer ambiguities.

Proof structure. Sections 2 and 3 establish the exact lattice, measure, and sign conventions. Sections 4–6 prove relative global localization. Sections 7–8 then prove the fixed-order expansion, and Sections 9 and 10 compute its first two corrections. The analytic estimates are independent of the finite checks in Section 12.

1.1 Provenance, sources and reading conventions

[write] **Added 6 October 2026, batch 108.** *Provenance.* This report is Report 194 of the session bundle that ProveIt’s docs/incoming intake processed as batch 108: the archive `A222959_Zero_Slope_Matrices_All_Orders_and_Inverses_Source.zip` (798,678 bytes, SHA-256 `87c6a152db72...a446a10`, 22 files at the archive root, no wrapper directory), arrival commit `60f54ea06`, placed in `602e5bd0f`. The text is the delivered `Report194.tex` (1613 lines, dated 4 October 2026), shipped as `article.tex`. The delivered 27-page PDF and the checksum manifest `SHA256SUMS.json` (21 entries, verified at placement) are not shipped, and the delivered README is replaced by the report README; all three are retrievable from the arrival commit. The package records no ProveIt commit (it cites nothing in the repository), no “prepared for private review” line and no personal data; its READMEs give generic `/tmp` example paths. It is a single-source report, so no merge choice was made. The write prefixed the 94 delivered labels with `zsm:` and updated the 83 references to them; it added this subsection, the notes marked [write], Remarks 11.1, 12.1 and 13.1, Proposition 14.1, the questions after the list of Section 14, the dated notes, and one bibliography entry (the b-files). No statement, proof or number of the source was changed: the added statements are the last numbered ones of their sections, the added subsection comes after the last text of Section 1, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (6 October 2026).

- OEIS A222959 [1] (revision #8, 3 October 2025) and A222955 [2] (revision #24, 17 March 2026), in the OEIS internal format, and both b-files (R. H. Hardin: A222959 by antidiagonals, $1 \leq n \leq 199$; A222955, $1 \leq n \leq 210$). Quoted in Remarks 12.1 and 13.1.
- Richomme [4], arXiv:2510.07159v1 (HTML): Theorems 5.13 and 5.14 and the paragraphs on A222955 and on the conjecture recorded there, read for Remark 13.1.
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:thm:lambert-core`, `p0:prop:factorial-core`, `p0:thm:core-reversion`, `p0:thm:flattening`, `p0:def:three-inverses`, `p0:thm:staircase` and `p0:rem:parity-instances`, read for Remark 11.1.

- Not read by the write: Prodinger [3] and the other works compared in Section 13 ([5, 6, 7, 8, 9, 10, 11]). What Section 13 says about them is the source’s account; `DATA_SOURCES.md` records what the source inspected. The one-row forms attributed to Prodinger there agree with the form Richomme states, $b_n \sim 2^{2\lfloor(n-1)/2\rfloor+1}(3/\pi)^{1/2}\lfloor n/2\rfloor^{-3/2}$.

What was checked at intake. The batch-108 dossier read the manuscript in full and found no error; it rechecked by hand $r_1 = 521/350$ and $r_2 = 2983/3500$, the parity ratio (6) and the two-step ratio (64), the smooth inverse (65) by constant-term algebra, the palindromic bound (69) at $n = 7, 8, 9$ and the energy and defect arithmetic of Section 6; its own meet-in-the-middle program gave $C_1, \dots, C_8$. It reran the package on a copy (the exact replay reproduces the delivered receipt byte for byte, normal and optimized; the guard tests pass; the optional $n = 9$ run gives 1808243216). The write recomputed every entry of the coefficient table of Section 10 from the displayed exact formulas for m , A , B , H_3 , $\sum_e h_e^4$, K and T (SymPy), and L_1 , L_2 , c_2 , r_1 , r_2 from them; rechecked the parity and two-step ratios and the smooth inverse through its X^{-1} term; recomputed the ratios 21.15, 3558.9, 23.66 of Section 12; counted C_n for $n \leq 10$ with a program of its own (Remark 12.1); and reran the package from a fresh extraction of the archive (note at the end of Section 12). Finite computations are observations, not certificates.

Relation to the repository. Before batch 108 no file of the repository mentioned A222959 or A222955, and none does outside this report and the collection’s catalogue entry for it. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. The nearest report in method is `a380592-tied-football-seasons`, in the same directory as this one (tied football seasons, A380592): a growing-dimensional Fourier count with its own global localization, every fixed order, and residue-class inverse thresholds. Its all-orders step imports Isaev’s complex cumulant theorem; the all-orders step here (Section 8) uses instead the one-sided bound $Z_K \geq -m(n)$ and a finite Taylor expansion, and imports no cumulant theorem. Neither report uses the other. The report `a005163-diagonally-symmetric-asms` is unrelated despite its matrix setting. The inversions of Section 11 are compared with the transseries volume in Remark 11.1.

[write] What is not claimed, collected from the source. Every fixed order only: no convergence, Borel summability or optimal truncation of the formal series, and no uniformity in an order J growing with n . No practical onset or numerical error bound: the localization constants (such as $\delta = 10^{-10}$) prove only the existence of a positive rate and an eventual threshold, “not a usable numerical error estimate”. The small exact counts are strongly preasymptotic and validate neither the error term nor an onset. The coefficients c_j , $j \geq 3$, are not computed, and the package is not an all-orders Wick engine. The envelopes (67) leave rounding ambiguities near threshold coincidences; no uniform single rounded smooth formula is claimed, and no interpolation is assigned the smooth formula’s X^{-1} coefficient. No absolute priority: the literature comparison is a focused search, “not a certificate of originality”; the obstructions to Barvinok–Hartigan concern the natural coordinates, not “every possible reformulation”; the Arratia–DeSalvo estimates are “conceptual predecessors”. The finite checks corroborate finite algebra and implementation only; they prove no localization, certify no threshold and are not a formal verification; reproducible PDF bytes are claimed only for a fixed TeX toolchain, and the integrity manifests are not signatures. The write adds: its Remark 13.1 rests on its own two-line proof and on Richomme’s text as read; its value of C_{10} is an exact computation reproducing the OEIS b-file, used by no proof.

Reading conventions (letters the manuscript reuses, with the tempting false readings; no symbol was renamed; T_{959} , ℓ_X and the subscript “vol” are the write’s).

Symbol	Meaning here (where)	Not to be read as
$C_n; T_{959}(n, k)$	the square diagonal of A222959, $C_1 = 2$; the whole array, n rows and k columns (Remark 12.1)	C the retained column set (Section 5), $C_G, C_{d,p}, C_K$; the colour C in Lemma 10.1
b_n, V_n	A222955(n): zero-slope words of length n , and the set of them	$V = \mathbb{E}D^2$ and the projection V of Section 9
$E, \mathbb{E}, \mathcal{E}, E_d$	phase subspace; expectation; modular energy; an inverse constant	the volume’s displacement E_{vol} (Remark 11.1)
d_0, d, D, D_d	AP spacing 1 or 2; $d \in \{12, 3\}$ with $S_n = n(n^2 - 1)/d$ (Section 11); $D = Q_4 - m(n)$; an inverse constant	the volume’s d of p0:prop:factorial-core
c, c_0, c_1, c_j	an absolute rate; the energy threshold; a Gaussian tail constant in (30); the coefficients of (4)	c the column margin vector of (7); $c_1 = 171/350$ is the coefficient
$h_{ij} = h_e, H$	projection diagonal (leverages); $\lfloor n/2 \rfloor$ in the lemmas, $\text{diag}(h)$, Q_8 , the hyperplane $(w, -w)^\perp$	the volume’s h_{vol}, H of p0:thm:flattening
K, F, T, L	target order of (36), $\text{Cov}(Q_4, Q_6)$, a kernel length, an integer matrix, K_J ; $ \Phi $, a Gaussian polynomial, Q_6 , lifts; $\mathbb{T}$, T_J , $\kappa_3(Q_4)$, T_0 ; margin lattice, $2J + 3$, $\log Y$, L_1, L_2	each section’s local letters, as the source warns in Section 10
$\kappa, \kappa(\cdot)$	the density constant 10^{-4} ; joint cumulants	the volume’s κ of p0:prop:factorial-core
$N(Y), X, x_\pm$	$\min\{n \geq 1 : C_n \geq Y\}$; $\sqrt{\log Y/A_0}$; envelope roots	the staircase N_* of p0:def:three-inverses , which presupposes a sequence increasing from some index on; X is also the volume’s core solution, here with the same meaning
u, v, ℓ_i, ℓ_X	an integer vector, centred weights, inverse corrections; normalized weights, defect count; loop counts; $\log X$ (the write’s)	

2 Lattices and Gaussian normalization

We work with Euclidean inner products, using the Frobenius inner product on matrix spaces. The covolume of a full lattice in a subspace means the Euclidean volume of one fundamental parallelepiped in that subspace.

2.1 The margin lattice

Let $A(M) = (Mw, M^T w)$. Its integer image is exactly

$$L = \{(r, c) \in \mathbb{Z}^{2n} : w^T r = w^T c\}. \quad (7)$$

Indeed, the compatibility equation is necessary. Choose $u \in \mathbb{Z}^n$ with $w^T u = 1$; one may choose a coordinate vector at a weight 1. For any compatible integer (r, c) , the integer matrix

$$M = ru^T + uc^T - (w^T r)uu^T \quad (8)$$

satisfies $Mw = r$ and $M^T w = c$. This proves saturation.

The primitive normal to the real margin subspace is $(w, -w)$, of length $\sqrt{2S}$, so $\text{covol}(L) = \sqrt{2S}$. To see the normal-vector formula, complete the primitive normal to a unimodular integer basis, or equivalently use a vector having integer dot product 1 with it. The corresponding height above its orthogonal hyperplane is the reciprocal of its length; unit ambient volume is that height times the hyperplane lattice covolume.

For independent Bernoulli entries of mean $1/2$, the mean margin vector is zero and its covariance on $H = (w, -w)^\perp$ is the restriction of

$$Q = \frac{1}{4} \begin{pmatrix} SI_n & ww^T \\ ww^T & SI_n \end{pmatrix}.$$

Its nonzero eigenvalues are $S/4$, of multiplicity $2n - 2$, and $S/2$, of multiplicity one. Thus

$$\det' Q = \frac{2S^{2n-1}}{4^{2n-1}}, \quad \frac{\text{covol}(L)}{(2\pi)^{(2n-1)/2} \sqrt{\det' Q}} = G_n.$$

This is an exact calculation of the Gaussian lattice factor, not by itself a local limit theorem.

2.2 A rectangular phase lattice lemma

The localization proof needs the same calculation after some rows and columns are removed. The retained weights need not be centered.

Lemma 2.1 (Primitive rectangular phase lattice). *Let $u \in \mathbb{Z}^r$ and $v \in \mathbb{Z}^s$ be nonzero primitive vectors, and write $S_u = \|u\|^2$, $S_v = \|v\|^2$. Set*

$$B(a, b) = av^T + ub^T, \quad E_{u,v} = \text{im}_{\mathbb{R}} B, \quad \Lambda_{u,v} = E_{u,v} \cap \mathbb{Z}^{r+s}.$$

Then $\dim E_{u,v} = r + s - 1$,

$$B(\mathbb{Z}^{r+s}) = \Lambda_{u,v}, \quad \text{covol}(\Lambda_{u,v}) = S_v^{(r-1)/2} S_u^{(s-1)/2}. \quad (9)$$

The map from $\mathbb{T}^{r+s}$ to $E_{u,v}/\Lambda_{u,v}$ is onto and pushes normalized Haar measure to normalized Haar measure. Its kernel is a single circle, with no additional finite components.

Proof. The real kernel is $\mathbb{R}(u, -v)$, which proves the dimension claim. Choose integer p, q with $u^T p = v^T q = 1$. For an integer $X \in E_{u,v}$, put

$$a = Xq, \quad b = X^T p - (p^T X q)v.$$

If $X = a_0 v^T + u b_0^T$, substitution gives $av^T + ub^T = X$; a, b are integer. This proves the first equality in (9).

The nonzero eigenvalues of $B^* B$ are S_v with multiplicity $r - 1$, S_u with multiplicity $s - 1$, and $S_u + S_v$ with multiplicity one. The integer kernel is generated by the primitive vector $(u, -v)$, of length $K = \sqrt{S_u + S_v}$. Projection of $\mathbb{Z}^{r+s}$ onto its orthogonal complement has covolume $1/K$: completing the kernel vector to a unimodular basis gives unit volume equal to K times the projected base volume. The Jacobian of B on that complement is

$$\sqrt{S_v^{r-1} S_u^{s-1} (S_u + S_v)}.$$

Multiplying by $1/K$ gives the claimed covolume.

Surjectivity on the real quotient is immediate, and the integer image calculation makes the torus map well-defined. An onto homomorphism of compact groups pushes normalized Haar measure to normalized Haar measure: its pushforward is a probability measure invariant under every target translation. Finally, if $B(a, b)$ is an integer matrix, saturation gives an integer pair (a', b') with the same image. Hence $(a - a', b - b')$ lies in the real kernel. Modulo integer coefficients, the kernel is exactly that real line modulo its primitive integer generator, a circle. $\square$

On any embedded real subset of $E_{u,v}/\Lambda_{u,v}$ the normalized Haar density is $1/\text{covol}(\Lambda_{u,v})$ times Euclidean volume. In particular,

$$\frac{1}{\text{covol}(\Lambda_{u,v})} \int_{E_{u,v}} e^{-\pi^2 \|X\|^2/2} dX = \left(\frac{2}{\pi}\right)^{(r+s-1)/2} S_v^{-(r-1)/2} S_u^{-(s-1)/2}. \quad (10)$$

For $u = v = w$, this is exactly G_n . The quotient calculation explains why no extra gauge-circle length belongs in the answer.

3 Fourier inversion and the positive central tube

The elementary identity $\int_{\mathbb{T}} e^{2\pi i k t} dt = 1_{\{k=0\}}$ yields

$$\frac{C_n}{2^{n^2}} = \int_{\mathbb{T}^{2n}} \prod_{i,j} \frac{1 + e^{2\pi i (w_j a_i + w_i b_j)}}{2} da db. \quad (11)$$

One may integrate all $2n$ margins even though one equation is redundant. For real representatives of the coefficient variables, write $X_{ij} = w_j a_i + w_i b_j$. Because the full weight lists are centered, $\sum_{ij} X_{ij} = 0$. Thus the signed characteristic function is

$$\Phi(a, b) = \prod_{ij} \cos(\pi X_{ij}), \quad |\Phi| = F. \quad (12)$$

Although one cosine is not a function on $\mathbb{R}/\mathbb{Z}$, its absolute value is, and the full signed product is well-defined. An integer change of a coefficient changes its sign by $(-1)^{\sum_j w_j} = 1$.

For $0 < r < 1/2$, let U_r consist of the phase-torus classes admitting a representative $X \in E$ with $\max |X_{ij}| < r$. Such a representative is unique when $2r < 1$: two representatives differ by an integer matrix with every entry of absolute value less than one. We use $r = 1/16$ and write $U = U_{1/16}$. If an arbitrary full real representative is $X + K$, with X in this tube and $K \in \Lambda$, then $\sum K_{ij} = 0$, since every matrix in E has total sum zero. Consequently

$$\prod \cos(\pi(X_{ij} + K_{ij})) = (-1)^{\sum K_{ij}} \prod \cos(\pi X_{ij}) = \prod \cos(\pi X_{ij}) > 0.$$

There is therefore no local sign cancellation or missing parity multiplier.

The only exact maximum phases of F are integer phases, which constitute the identity of E/Λ ; Lemma 2.1 identifies their coefficient preimage as the gauge circle $(a, b) = (tw, -tw)$. This fact alone does not exclude nearby or distant approximate resonances. The next three sections establish the necessary quantitative statement.

We will use the global scalar estimate

$$|\cos(\pi t)| \leq \exp\left(-\frac{\pi^2}{2} \|t\|_{\mathbb{T}}^2\right), \quad \|t\|_{\mathbb{T}} = \text{dist}(t, \mathbb{Z}). \quad (13)$$

For $|x| < 1/2$, the derivative of $\log \cos(\pi x) + \pi^2 x^2/2$ is $-\pi \tan(\pi x) + \pi^2 x$, nonpositive for $x \geq 0$; symmetry and the endpoint limit prove the inequality. Outside the central tube, the modular distance in (13) cannot be replaced by an arbitrary real lift.

4 Dense arithmetic progression inverse lemmas

All statements in this section hold for sufficiently large n , uniformly in its parity. Their deliberately generous constants make the required density and no-wrap conditions transparent.

Lemma 4.1 (Scalar inverse). *Suppose $0 < \tau < 1/8$ and $0 \leq \rho < 1/4$. If $t \in \mathbb{T}$ satisfies $\|w_j t\|_{\mathbb{T}} \leq \tau$ for at least $(1 - \rho)n$ indices, then*

$$\|t\|_{\mathbb{T}} \leq \frac{8\tau}{n}.$$

Proof. Let $H = \lfloor n/2 \rfloor$. For each $1 \leq h \leq H$, at least one pair $j, j+h$ has both endpoints good: there are $n-h$ pairs and at most $2\rho n$ are spoiled. Subtraction gives $\|d_0 h t\|_{\mathbb{T}} \leq 2\tau$ for every such h .

We first note an elementary no-wrap fact. If $\|hz\|_{\mathbb{T}} \leq A < 1/4$ for $1 \leq h \leq H$, then $\|z\|_{\mathbb{T}} \leq A/H$. Otherwise, since $\|z\|_{\mathbb{T}} \leq A$, the first h with $h\|z\|_{\mathbb{T}} > A$ satisfies $A < h\|z\|_{\mathbb{T}} \leq 2A < 1/2$. At this first crossing there is no wrap, contradicting $\|hz\|_{\mathbb{T}} \leq A$.

Apply this fact with $z = d_0 t$ and $A = 2\tau$. If $d_0 = 1$, it gives the result. If $d_0 = 2$, then t is within τ/H of an integer or a half-integer. The half-integer branch is impossible: every weight is odd, and any good weight would have distance at least $1/2 - n\tau/H \geq 1/2 - 3\tau > \tau$ from an integer. The integer branch gives the claimed bound. The constants cover all rounding for sufficiently large n . $\square$

Lemma 4.2 (Dense circuit stability). *Put $\kappa = 10^{-4}$. Let $I \subseteq \{1, \dots, n\}$ omit at most κn indices. Suppose $\theta_i \in \mathbb{T}$, $i \in I$, and*

$$\left\| \sum_i z_i \theta_i \right\|_{\mathbb{T}} \leq \eta \tag{14}$$

for every integer vector z supported in I with $\sum_i z_i w_i = 0$ and $\sum_i |z_i| \leq 4$. If $6n\eta < 1/2$, there is $t \in \mathbb{T}$ such that

$$\|\theta_i - w_i t\|_{\mathbb{T}} \leq 500\eta \quad (i \in I). \tag{15}$$

Proof. Set $H = \lfloor n/4 \rfloor$. For each $0 \leq h \leq H$, choose a pair $x, x+h \in I$ and define $f(h) = \theta_{x+h} - \theta_x$, with $f(0) = 0$. Pairs exist because $n-h-2\kappa n > 0$. Any two choices differ in circle norm by at most η : their difference is a four-term circuit in (14). For $h, k \geq 0$, $h+k \leq H$, a triple $x, x+h, x+h+k \in I$ exists because at least $n-H-3\kappa n > 0$ starting positions remain. Comparing the three chosen increments gives

$$\|f(h+k) - f(h) - f(k)\|_{\mathbb{T}} \leq 3\eta. \tag{16}$$

Choose $F(0) = 0$, choose any real lift $F(1)$, and choose subsequent real lifts recursively with $|F(h) - F(h-1) - F(1)| \leq 3\eta$. Then $|F(h) - hF(1)| \leq 3(h-1)\eta$. For positive h, k with $h+k \leq H$, the real additive defect has absolute value at most $6H\eta < 1/2$. It is a lift of the circle element in (16), so its absolute value is at most 3η . This step is what prevents an accumulated integer winding error.

Put $A = F(H)/H$ and $g(h) = F(h) - Ah$. For $2h \leq H$, $|2g(h) - g(2h)| \leq 3\eta$. For $2h > H$, add the additive relations for $H = h + (H-h)$ and $h = (H-h) + (2h-H)$ to obtain $|2g(h) - g(2h-H)| \leq 6\eta$. Since $g(0) = g(H) = 0$, the maximum $M = \max_{0 \leq h \leq H} |g(h)|$ satisfies $M \leq M/2 + 3\eta$, hence $M \leq 6\eta$. Every actual pair in I at distance at most H consequently satisfies

$$\|\theta_y - \theta_x - A(y-x)\|_{\mathbb{T}} \leq 7\eta.$$

Partition the full index interval into sixteen consecutive blocks of nearly equal length. Every block contains an element of I . For sufficiently large n , adjacent selected representatives are at distance

at most H , and any point can be joined to one fixed representative by at most sixteen such links. Thus, for some real b ,

$$\|\theta_i - Ai - b\|_{\mathbb{T}} \leq 112\eta \quad (i \in I).$$

Recenter the progression: in circle notation $\theta_i = (A/d_0)w_i + B + e_i$, with real lifts $|e_i| \leq 112\eta$.

If $n = 2m + 1$ is odd, view the retained weights as a dense subset of $[-m, m] \cap \mathbb{Z}$. Choose a retained p in $[-\lfloor m/4 \rfloor, \lfloor m/4 \rfloor]$. Among the possible q in $[-\lfloor m/2 \rfloor, \lfloor m/2 \rfloor]$, at most $2\kappa n$ fail one of the conditions that q and $p + q$ be retained. All sums are inside $[-m, m]$, so a retained triple $p, q, r = p + q$ exists. Its three-term circuit gives $\|B\|_{\mathbb{T}} \leq \eta + 3(112\eta) = 337\eta$. Repeated indices simply combine coefficients and cause no problem. Hence the total error is at most $449\eta < 500\eta$.

If n is even, a retained opposite-weight pair exists, since otherwise at least $n/2$ weights would be omitted. It gives $\|2B\|_{\mathbb{T}} \leq \eta + 2(112\eta) = 225\eta$. Thus B is within 112.5η of an integer or a half-integer. All w_i are odd, so a half-integer constant is absorbed into the coefficient of w_i modulo one. The total error is then at most $224.5\eta < 500\eta$. This proves (15) in both parities. $\square$

5 A robust central core and its Gaussian cost

Fix, once and for all,

$$\epsilon = 10^{-3}, \quad \kappa = 10^{-4}, \quad \delta = 10^{-10}. \quad (17)$$

An entry is called bad when $\|x_{ij}\|_{\mathbb{T}} \geq \delta$.

Lemma 5.1 (Robust central core). *Let $R, C \subseteq \{1, \dots, n\}$ each omit at most κn indices. Suppose every row in R and every column in C has at most ϵn bad entries in its full row or column. There is $t \in \mathbb{T}$ such that*

$$\|a_i - w_i t\|_{\mathbb{T}} \leq 16000\delta/n \quad (i \in R), \quad (18)$$

$$\|b_j + w_j t\|_{\mathbb{T}} \leq 2064064\delta/n \quad (j \in C). \quad (19)$$

The core phases have a real lift in E_{w_R, w_C} with every entry of absolute value at most $2080064\delta < 1/16$.

Proof. For any integer circuit z supported in R with ℓ^1 norm at most four, at most $4\epsilon n$ full columns meet a bad entry in its support. In every other column,

$$\|w_j \sum_i z_i a_i\|_{\mathbb{T}} = \|\sum_i z_i x_{ij}\|_{\mathbb{T}} \leq 4\delta.$$

Lemma 4.1, with $\rho = 4\epsilon < 1/4$ and $\tau = 4\delta$, gives $\|\sum_i z_i a_i\|_{\mathbb{T}} \leq 32\delta/n$. Apply Lemma 4.2 with $\eta = 32\delta/n$; its lift condition follows from $192\delta < 1/2$. We get $a_i = w_i t + e_i$ modulo one, where $|e_i| \leq 16000\delta/n$. By the transposed argument $b_j = w_j s + f_j$, with $|f_j| \leq 16000\delta/n$.

Put $h = t + s$. On a good core entry, using $|w_i| \leq n$,

$$\|w_i w_j h\|_{\mathbb{T}} \leq \delta + |w_j e_i| + |w_i f_j| \leq 32001\delta =: \tau.$$

For each retained row more than $3n/4$ retained columns are good. Scalar inversion in j gives $\|w_i h\|_{\mathbb{T}} \leq 8\tau/n$ for every $i \in R$. The retained row set is dense, so a second scalar inversion gives

$$\|h\|_{\mathbb{T}} \leq 64\tau/n^2 = 2048064\delta/n^2.$$

Choose the nearest-integer real lift of h and combine it with f_j . This gives (19), while (18) already holds. Using these real residual coefficients, the lift is $e_i w_j + w_i \tilde{f}_j$; it genuinely lies in the core phase space and its maximum is at most $(16000 + 2064064)\delta = 0.0002080064 < 1/16$. $\square$

For retained sizes $r = |R|$, $s = |C|$, write

$$S_R = \sum_{i \in R} w_i^2, \quad S_C = \sum_{j \in C} w_j^2, \quad G_{R,C} = \left(\frac{2}{\pi}\right)^{(r+s-1)/2} S_C^{-(r-1)/2} S_R^{-(s-1)/2}.$$

A dense retained list contains consecutive indices. For odd n , the corresponding weights are consecutive integers; for even n , they are consecutive odd numbers with difference two. Their greatest common divisor is one in either case. Thus both retained lists are primitive and Lemma 2.1 applies.

On the embedded core tube, (13) and (10) give

$$\int_{\{\text{core in its central tube}\}} \prod_{i \in R, j \in C} |\cos(\pi x_{ij})| da_R db_C \leq G_{R,C}. \quad (20)$$

Only an absolute-product estimate is used here. Restricted weights need not sum to zero, so we do not assert the signed-integrand property for a core.

Lemma 5.2 (Uniform Gaussian cost per defect). *Let $k = n - r$, $l = n - s$, and $v = k + l \leq \kappa n$. Then*

$$\frac{G_{R,C}}{G_n} \leq (C_G n^{3/2})^v, \quad C_G = \sqrt{\pi/2} e^{16}. \quad (21)$$

Proof. Uniformly in both parities, $n^3/16 \leq S \leq n^3$ for $n \geq 2$, and $\max_i w_i^2 \leq n^2$. Therefore

$$S_R/S \geq 1 - 16k/n, \quad S_C/S \geq 1 - 16l/n,$$

and, as $16\kappa < 1/2$,

$$\log(S/S_R) \leq 32k/n, \quad \log(S/S_C) \leq 32l/n.$$

The exact ratio is

$$\frac{G_{R,C}}{G_n} = \left(\frac{\pi}{2}\right)^{v/2} S^{v/2} \left(\frac{S}{S_C}\right)^{(r-1)/2} \left(\frac{S}{S_R}\right)^{(s-1)/2}.$$

Bounding the last two logarithms by $16l$ and $16k$ proves the claim. In particular there is no extraneous factor $e^{O(n)}$ independent of v . $\square$

6 Global relative localization by defects

We now prove Theorem 1.2. Define the modular energy

$$\mathcal{E}(a, b) = \sum_{i,j} \|x_{ij}\|_{\mathbb{T}}^2, \quad c_0 = \delta^2 \epsilon \kappa / 2.$$

On $\mathcal{E} > c_0 n^2$, (13) gives the integral bound

$$\int_{\{\mathcal{E} > c_0 n^2\}} F \leq e^{-\pi^2 c_0 n^2 / 2} = O(G_n e^{-cn}) \quad (22)$$

for some $c > 0$ and all sufficiently large n , because $\log G_n = -3n \log n + O(n)$. Comparison with G_n is essential here.

On the low-energy region, remove simultaneously every row and every column with more than ϵn bad entries, counting bad entries in the original full rows and columns. Let k, l be the numbers

removed, $v = k + l$, and let R, C be the retained sets. If B is the total number of bad entries, then $B\delta^2 \leq \mathcal{E}$, and

$$v \leq \frac{2B}{\epsilon n} \leq \frac{2\mathcal{E}}{\delta^2 \epsilon n} \leq \kappa n. \quad (23)$$

Thus the retained core satisfies Lemma 5.1.

Each removed row has at least $\epsilon n - l \geq \epsilon n/2$ bad entries in retained columns. Each removed column likewise has at least $\epsilon n/2$ bad entries in retained rows, since $\kappa \leq \epsilon/2$. These charges are on the disjoint sets $R^c \times C$ and $R \times C^c$, and no charged factor is counted twice. There are at least $\epsilon n v/2$ charged factors, so

$$F \leq e^{-\pi^2 \delta^2 \epsilon n v/4} \prod_{i \in R, j \in C} |\cos(\pi x_{ij})|. \quad (24)$$

This analytic form avoids evaluating $\cos(\pi\delta)$ in floating point, where it would round to one.

Fix particular removed sets, and let $\mathcal{A}_{R,C}$ be the event that these are exactly the removed sets and that $\mathcal{E} \leq c_0 n^2$. Put $\gamma = \pi^2 \delta^2 \epsilon/4 > 0$. On this event the core lies in its central tube. Enlarge just that event after applying (24):

$$\begin{aligned} \int 1_{\mathcal{A}_{R,C}} F da db &\leq e^{-\gamma n v} \int 1_{\{\text{core in central tube}\}} \prod_{i \in R, j \in C} |\cos(\pi x_{ij})| da db \\ &\leq e^{-\gamma n v} G_{R,C}. \end{aligned} \quad (25)$$

The discarded coefficient coordinates integrate over full tori of total volume one. The remaining variables push forward to core Haar measure as in Lemma 2.1. This argument does not assume independence of the defect event and does not introduce a conditional fiber-volume loss.

For each v , the number of pairs of removed sets is $\sum_{k+l=v} \binom{n}{k} \binom{n}{l} = \binom{2n}{v} \leq (2n)^v$. By (21), the total contribution of nonempty low-energy defect patterns is at most

$$G_n \sum_{v \geq 1} \left(2C_G n^{5/2} e^{-\gamma n} \right)^v = O(G_n e^{-c n}). \quad (26)$$

The last step holds eventually, because the bracket is at most $e^{-\gamma n/2}$ for all sufficiently large n . If $v = 0$, Lemma 5.1 applies to the whole phase matrix and puts it in U . Hence every point outside U is covered by (22) or (26). This proves Theorem 1.2.

What the constants do and do not establish. The small fixed δ proves existence of a positive localization rate and an eventual threshold. It is not a usable numerical error estimate. The cost of one defect is only polynomial in n , whereas its cross-edge penalty is exponential in n ; that comparison is the structural reason the proof closes.

7 The local Gaussian and a dimension free moment bound

7.1 Projection and shrinking cutoff

Put $v_i = w_i/\sqrt{S}$, $a_i = v_i^2$, and $s_r = \sum_i a_i^r$. The orthogonal projection onto E is

$$P(X) = X v v^T + v v^T X - v v^T X v v^T. \quad (27)$$

Indeed the two first summands are commuting orthogonal projections whose intersection is spanned by $v v^T$. In entry notation,

$$P_{(ij),(kl)} = 1_{\{i=k\}} v_j v_l + 1_{\{j=l\}} v_i v_k - v_i v_j v_k v_l. \quad (28)$$

Its diagonal entries are

$$h_{ij} = a_i + a_j - a_i a_j, \quad \max h_{ij} = O(n^{-1}), \quad \sum_{ij} h_{ij} = 2n - 1. \quad (29)$$

Here $s_1 = 1$ and $\max a_i = O(n^{-1})$, directly from (2).

Under the normalized Gaussian with density proportional to $e^{-\pi^2 \|X\|^2/2}$ on E , the matrix $Y = \pi X$ is a centered Gaussian with covariance P . Fix any $0 < \zeta < 1/6$ and let

$$\mathcal{B}_n = \{\max_{ij} |Y_{ij}| \leq n^{-1/2+\zeta}\}.$$

The Gaussian one-coordinate tail and a union bound give

$$\mathbb{P}(\mathcal{B}_n^c) \leq 2n^2 e^{-c_1 n^{2\zeta}}. \quad (30)$$

For completeness, the scalar bound follows by applying Markov's inequality to e^{tZ} , using $\mathbb{E}e^{tZ} = e^{t^2 \sigma^2/2}$, and minimizing over t ; apply it to both signs. The cutoff tube lies in U for sufficiently large n . On the intervening part of U , the absolute integrand is bounded by its Gaussian quadratic part. Global localization, the positive sign in U , and (30) therefore imply, for every fixed M ,

$$\frac{C_n}{2n^2 G_n} = \mathbb{E} \left[1_{\mathcal{B}_n} \exp \left\{ \sum_{ij} \left(\log \cos Y_{ij} + \frac{Y_{ij}^2}{2} \right) \right\} \right] + O_M(n^{-M}). \quad (31)$$

All logs inside this cutoff are of positive cosines.

7.2 A self contained Gaussian polynomial estimate

We record the dimension-free estimate required to expand the quartic term. For a random variable Z , write $\|Z\|_p = (\mathbb{E}|Z|^p)^{1/p}$.

Lemma 7.1 (Fixed-degree Gaussian moments). *For each fixed nonnegative integer d and $1 \leq p < \infty$, there is a constant $C_{d,p}$ independent of the dimension such that every polynomial $F(g)$ of degree at most d in a standard Gaussian vector satisfies*

$$\|F\|_p \leq C_{d,p} \|F\|_2.$$

Proof. Choose an even integer $2m \geq p$; norm monotonicity reduces the claim to $2m$. Define the multivariate Wick polynomials by the generating identity

$$e^{t^T g - \|t\|^2/2} = \sum_{k \geq 0} \frac{1}{k!} \sum_{i_1, \dots, i_k} t_{i_1} \cdots t_{i_k} : g_{i_1} \cdots g_{i_k} :.$$

Taking the expectation of products of these generating functions yields $\exp(\sum_{a < b} t_a^T t_b)$. Coefficient comparison proves orthogonality of different total degrees and Wick's rule: pairings join slots in different Wick-ordered factors only. Since these polynomials have the ordinary monomials as leading terms, every F has an orthogonal finite decomposition $F = \sum_{k=0}^d F_k$, where

$$F_k = \sum_{i_1, \dots, i_k} T_{i_1 \dots i_k} : g_{i_1} \cdots g_{i_k} :$$

for a symmetric tensor T , and $\|F_k\|_2^2 = k! \|T\|_{\text{HS}}^2$.

For $k \geq 1$, the moment $\mathbb{E}F_k^{2m}$ is a sum over at most $(2mk - 1)!!$ pairings of slots, with no pairing internal to one factor. Each pairing is a tensor contraction along a loop-free multigraph with $2m$ tensor vertices. Its absolute value is at most $\|T\|_{\text{HS}}^{2m}$. To prove this last claim, if tensors A, B share index tuple I and have disjoint remaining tuples J, K , form $C_{J,K} = \sum_I A_{I,J} B_{I,K}$. Cauchy–Schwarz gives

$$\sum_{J,K} |C_{J,K}|^2 \leq \left(\sum_{I,J} |A_{I,J}|^2 \right) \left(\sum_{I,K} |B_{I,K}|^2 \right).$$

Merge two connected tensor blocks and contract all their common edges. Continue in each connected component; all newly internal edges are removed at each merge, so no uncontrolled self-trace appears. Disconnected components multiply. This proves the contraction bound and hence

$$\|F_k\|_{2m} \leq \left(\frac{(2mk - 1)!!}{(k!)^m} \right)^{1/(2m)} \|F_k\|_2.$$

The $k = 0$ case is immediate. Minkowski followed by Cauchy–Schwarz over the fixed set $0 \leq k \leq d$ and orthogonality proves the claim, with no ambient dimension in the constant. $\square$

7.3 Quartic concentration and higher powers

Define positive rational coefficients by

$$-\log \cos z = \frac{z^2}{2} + \sum_{k \geq 2} q_k z^{2k}, \quad q_2 = \frac{1}{12}, \quad q_3 = \frac{1}{45}, \quad q_4 = \frac{17}{2520},$$

and put $Q_{2k} = q_k \sum_{ij} Y_{ij}^{2k}$. Rationality and positivity follow recursively from $\tan' = 1 + \tan^2$, $\tan 0 = 0$, followed by integration. For each fixed k, p , Minkowski and scalar Gaussian moments give

$$\|Q_{2k}\|_p \leq C_{k,p} \sum_e h_e^k \leq C_{k,p} (\max h)^{k-1} \text{tr } P = O_{k,p}(n^{2-k}), \quad (32)$$

where a single index e denotes a cell.

The quartic mean is

$$m(n) = \mathbb{E}Q_4 = \frac{1}{4} \sum_e h_e^2 = \frac{2ns_2 + 2 - 4s_2 + s_2^2}{4}, \quad s_2 = \frac{3(3n^2 - 7)}{5n(n^2 - 1)}. \quad (33)$$

In particular $m(n) = 7/5 + O(n^{-1})$. For jointly centered Gaussian variables, Wick pairing gives

$$\text{Cov}(Y_e^4, Y_f^4) = 72h_e h_f P_{ef}^2 + 24P_{ef}^4.$$

There are 72 pairings with exactly two cross pairs and 24 with four; the no-cross pairings cancel in the covariance. Thus

$$\text{Var } Q_4 = \frac{1}{2} \sum_{e,f} h_e h_f P_{ef}^2 + \frac{1}{6} \sum_{e,f} P_{ef}^4 = O(n^{-1}). \quad (34)$$

Indeed $|P_{ef}|^2 \leq h_e h_f$, by positivity of a covariance matrix, and $\sum_{e,f} P_{ef}^2 = \text{tr } P = 2n - 1$. Each displayed sum is bounded by a constant times $(\max h)^2 \text{tr } P$. For $D = Q_4 - m(n)$, Lemma 7.1 consequently yields

$$\|D\|_p = O_p(n^{-1/2}) \quad (35)$$

for every fixed p . This remains valid in the growing dimension $2n - 1$.

8 A constructive expansion to every fixed order

We now prove Theorem 1.1, except for the evaluations of c_1 and c_2 .

8.1 Two finite Taylor expansions with controlled remainders

Fix $K \geq 2$. On $|y| \leq 1/2$, positivity and analyticity of the log-cosine series imply

$$0 \leq -\log \cos y - y^2/2 - \sum_{k=2}^K q_k y^{2k} \leq C_K |y|^{2K+2}.$$

Hence on $\mathcal{B}_n$, for all sufficiently large n ,

$$\sum_e (\log \cos Y_e + Y_e^2/2) = - \sum_{k=2}^K Q_{2k} - R_K, \quad 0 \leq R_K \leq C_K \sum_e |Y_e|^{2K+2}.$$

The difference of the two exponentials with and without R_K is at most R_K , since the retained Q 's are nonnegative. Its expected value is $O_K(\sum_e h_e^{K+1}) = O_K(n^{1-K})$. Removing the cutoff from $\exp(-\sum Q_{2k}) \leq 1$ costs at most (30). Therefore

$$\frac{C_n}{2^{n^2} G_n} = \mathbb{E} \exp \left(- \sum_{k=2}^K Q_{2k} \right) + O_K(n^{1-K}). \quad (36)$$

No pointwise maximum bound multiplied by n^2 is used for this remainder; the integrated Gaussian moments provide its stated order.

For the target order $J \geq 0$, choose

$$K = J + 3, \quad L = 2J + 3, \quad Z_K = D + \sum_{k=3}^K Q_{2k}. \quad (37)$$

Equations (32) and (35) show $\|Z_K\|_p = O_{J,p}(n^{-1/2})$. Crucially, $Z_K = \sum_{k=2}^K Q_{2k} - m(n) \geq -m(n)$. All derivatives of e^{-z} along the segment from 0 to Z_K are bounded by $e^{m(n)} = O(1)$. The ordinary finite Taylor theorem, for every realization, thus gives

$$\left| e^{-Z_K} - \sum_{r=0}^L \frac{(-Z_K)^r}{r!} \right| \leq \frac{e^{m(n)}}{(L+1)!} |Z_K|^{L+1}.$$

Its expectation is $O_J(n^{-J-2})$, as is the error in (36). Define the finite quantity

$$T_J(n) = \sum_{r=0}^{2J+3} \frac{(-1)^r}{r!} \mathbb{E} Z_{J+3}^r. \quad (38)$$

We have proved the stronger intermediate approximation

$$\frac{C_n}{2^{n^2} G_n} = e^{-m(n)} T_J(n) + O_J(n^{-J-2}). \quad (39)$$

There has been no exchange of an infinite series with an expectation. The one-sided bound on Z_K avoids any need for an exponential moment of $|D|$.

The polynomial families in this argument also have negligible weighted tails on $\mathcal{B}_n^c$: Hölder and the fixed-moment bounds give polynomial L^2 bounds for every fixed product, and Cauchy–Schwarz with (30) makes the tail superpolynomially small. This assertion is about these controlled families, not arbitrary polynomials with unrestricted n -dependent coefficients.

8.2 Exact rationality by Wick pairings and power sums

For this part use $u_i = i - (n + 1)/2$ for every n , even when the u_i are half-integers, and put $S_u = n(n^2 - 1)/12$. Scaling all weights by a nonzero constant does not change E or P . Thus in both parities the same covariance formula is

$$P_{(ij),(kl)} = 1_{\{i=k\}} \frac{u_j u_l}{S_u} + 1_{\{j=l\}} \frac{u_i u_k}{S_u} - \frac{u_i u_j u_k u_l}{S_u^2}. \quad (40)$$

Every fixed Wick moment in (38) is an exact rational function of n . Here is a constructive reduction proving that claim.

1. Expand the finitely many powers of $D = Q_4 - m(n)$ and the higher Q 's. All entry-index sums initially run without restrictions; $m(n)$ is already rational.
2. Apply Wick's rule to each Gaussian monomial, a finite sum of covariance products. The number of slots and pairings depends only on J .
3. For each covariance factor choose one of the three terms in (40). Merge index variables wherever a Kronecker delta imposes equality.
4. Sum each remaining free index independently. Every result is a rational coefficient times a power of S_u^{-1} and a product of centered power sums

$$M_r(n) = \sum_{i=1}^n \left(i - \frac{n+1}{2} \right)^r, \quad r \geq 0. \quad (41)$$

Each $M_r(n)$ is a polynomial with rational coefficients. One elementary power-sum recursion is

$$\sum_{i=1}^n i^r = \frac{(n+1)^{r+1} - 1 - \sum_{k=0}^{r-1} \binom{r+1}{k} \sum_{i=1}^n i^k}{r+1},$$

which follows by telescoping $(i+1)^{r+1} - i^{r+1}$. Binomially expand the centered powers in (41); odd centered powers vanish by reflection, and $M_0(n) = n$. This supplies a finite rational implementation without invoking an unevaluated summation theorem.

Repeated entry indices cause no missing case. They are present in the initial unrestricted sums, and the delta equalities only merge variables. No condition that surviving indices be distinct occurs, so no hidden inclusion-exclusion is required. This proves exact rationality and parity independence of T_J . Primitive integral weights remain essential for the lattice factor G_n ; they are not required for the covariance calculation.

8.3 Coefficient extraction and consistency

Because $\mathbb{E}D = 0$, $\mathbb{E}Z_K = \sum_{k=3}^K \mathbb{E}Q_{2k} = O(n^{-1})$. For every fixed $r \geq 2$, $|\mathbb{E}Z_K^r| \leq \|Z_K\|_r^r = O(n^{-r/2})$. Hence

$$T_J(n) = 1 + O(n^{-1}).$$

As a rational function, T_J therefore has no pole at infinity: its Laurent expansion contains only a constant and negative powers of n . Likewise $m(n) - 7/5 = O(n^{-1})$ has rational coefficients there. For $0 \leq j \leq J$, define

$$c_j = [z^j] \{ \exp(7/5 - m(1/z)) T_J(1/z) \}. \quad (42)$$

The expression in braces is analytic near $z = 0$, has constant term one, and has rational Taylor coefficients. Its finite Taylor remainder through degree J is $O_J(n^{-J-1})$ at $z = 1/n$. Together with (39), this proves (4).

If two target orders J, J' are used, both finite analytic approximants approximate the same normalized exact count. Their difference along $z = 1/n$ is $O(n^{-\min(J, J')-2})$. An analytic function with a first nonzero term of lower degree could not have that bound. Their coefficients therefore agree through every common displayed order. Thus (42) defines one consistent sequence (c_j) .

This is a constructive all-fixed-orders result, not an efficient algorithm claim: naive Wick enumeration is expensive. It does not imply that the full formal series converges, that J may grow with n , or that truncation errors are numerically small at the currently tabulated dimensions.

9 The explicit first correction

We give enough exact algebra to isolate the value $171/350$ and to make its independent rational checks transparent. Finite centered power summation gives

$$s_2 = \frac{3(3n^2 - 7)}{5n(n^2 - 1)} = \frac{9}{5n} + O(n^{-3}), \quad (43)$$

$$s_3 = \frac{9(3n^4 - 18n^2 + 31)}{7n^2(n^2 - 1)^2} = \frac{27}{7n^2} + O(n^{-4}). \quad (44)$$

These identities hold exactly for both parities because normalized weights are scale-invariant. Equation (33) gives

$$m(n) = \frac{7}{5} - \frac{9}{5n} + O(n^{-2}). \quad (45)$$

Expanding $(a_i + a_j - a_i a_j)^3$ and summing yields

$$\begin{aligned} H_3 &:= \sum_{ij} h_{ij}^3 = 2ns_3 + 6s_2 - 6s_3 - 6s_2^2 + 6s_2s_3 - s_3^2 \\ &= \frac{648}{35n} + O(n^{-2}). \end{aligned} \quad (46)$$

Since $\mathbb{E}Y_e^6 = 15h_e^3$,

$$\mathbb{E}Q_6 = \frac{1}{3}H_3 = \frac{216}{35n} + O(n^{-2}). \quad (47)$$

9.1 The two variance sums

Write

$$A = \sum_{e,f} h_e h_f P_{ef}^2, \quad B = \sum_{e,f} P_{ef}^4.$$

Let U, V be respectively the same-row and same-column projections in (28), and let $R = zz^T$ with $z = v \otimes v$. Then $P = U + V - R$ and $A = \text{tr}(H P H P)$ for $H = \text{diag}(h)$. The following intermediate

finite sums evaluate its trace terms:

$$\begin{aligned}
F_0 &= \sum_i [a_i + (1 - a_i)s_2]^2 = 3s_2 + (n - 4)s_2^2 + s_2^3, \\
T_0 &= \sum_{ij} h_{ij}^2 a_i a_j = 2s_3 + s_3^2 + 2s_2^2 - 4s_2 s_3, \\
W_0 &= \sum_i a_i [a_i + (1 - a_i)s_2]^2 = (1 - s_2)^2 s_3 + 2s_2^2(1 - s_2) + s_2^2, \\
V_0 &= \sum_{ij} h_{ij} a_i a_j = 2s_2 - s_2^2.
\end{aligned}$$

Expansion of the trace gives $A = 2F_0 + 2T_0 - 4W_0 + V_0^2$. Hence the exact identity

$$A = 6s_2 + (2n - 12)s_2^2 + 6s_2^3 + s_2^4 - 4s_2^2 s_3 + 2s_3^2 = \frac{432}{25n} + O(n^{-2}). \quad (48)$$

For B , divide the ordered pair of cells into four disjoint cases: same row only, same column only, equal cells, and different rows and columns. With $s_4 = \sum_i a_i^4$, this gives

$$B = 2 \left[\sum_i (1 - a_i)^4 \right] (s_2^2 - s_4) + \sum_{ij} h_{ij}^4 + (s_2^2 - s_4)^2. \quad (49)$$

For example, on the same-row-only case, $P_{(ij),(il)} = (1 - a_i)v_j v_l$, and the sum over $j \neq l$ contributes $s_2^2 - s_4$. Direct expansion gives

$$\begin{aligned}
\sum_{ij} h_{ij}^4 &= 6s_2^2 - 24s_2 s_3 + 12s_2 s_4 + 12s_3^2 - 8s_3 s_4 \\
&\quad + 8s_3 + s_4^2 + 2ns_4 - 8s_4.
\end{aligned}$$

After substitution into (49), every s_4 term cancels:

$$\begin{aligned}
B &= (2n - 2)s_2^2 + 12s_3^2 + s_4^2 + 8s_3 - 24s_2 s_3 - 8s_2^2 s_3 + 12s_3^2 \\
&= \frac{162}{25n} + O(n^{-2}).
\end{aligned} \quad (50)$$

By (34),

$$\text{Var } Q_4 = \frac{1}{2}A + \frac{1}{6}B = \frac{243}{25n} + O(n^{-2}). \quad (51)$$

9.2 Remainder control and the coefficient

Taking $K = 3$ in (36) gives an $O(n^{-2})$ error. Since $D \geq -m$ and $m = O(1)$, Taylor's theorem and (35) yield

$$\mathbb{E}e^{-Q_4} = e^{-m} \left(1 + \frac{1}{2} \text{Var } Q_4 + O(n^{-3/2}) \right).$$

Moreover $\|Q_6\|_2 = O(n^{-1})$ and e^{-x} is 1-Lipschitz on $[0, \infty)$, so

$$\left| \mathbb{E}[(e^{-Q_4} - e^{-m})Q_6] \right| \leq \|D\|_2 \|Q_6\|_2 = O(n^{-3/2}).$$

Using $|e^{-x} - 1 + x| \leq x^2/2$ for $x \geq 0$, we get

$$\mathbb{E}e^{-Q_4 - Q_6} = e^{-m} \left(1 + \frac{1}{2} \text{Var } Q_4 - \mathbb{E}Q_6 + O(n^{-3/2}) \right).$$

The order n^{-1} correction inside these parentheses is

$$\frac{243}{50} - \frac{216}{35} = -\frac{459}{350}.$$

Multiplication by (45) contributes $9/5$, giving

$$c_1 = \frac{9}{5} - \frac{459}{350} = \frac{171}{350}.$$

This calculation on its own has an $O(n^{-3/2})$ remainder. The all-fixed-orders theorem already proved in Section 8, and uniqueness of the first coefficient, improve the first-order statement to $O(n^{-2})$. No uncomputed higher coefficient is inferred from this remainder improvement.

10 The explicit second correction

The finite coefficient prescription can be evaluated one order further. We prove

$$c_2 = \frac{483051}{245000}, \quad \log \left(\frac{e^{7/5} C_n}{2^{n^2} G_n} \right) = \frac{171}{350n} + \frac{6483}{3500n^2} + O(n^{-3}). \quad (52)$$

No coefficient is fitted to finite counts. The additional ingredient is a connected-Wick power bound, followed by two finite contractions.

10.1 Connected Wick pairings and their power count

For random variables with moments, joint cumulants are defined recursively by

$$\mathbb{E} \prod_{i=1}^r Z_i = \sum_{\mathcal{P}} \prod_{B \in \mathcal{P}} \kappa(Z_i : i \in B),$$

where the sum is over all set partitions of $\{1, \dots, r\}$. This is a finite algebraic definition and does not assume existence of a moment-generating function near zero.

Lemma 10.1 (Connected power count). *For fixed integers $k_i \geq 2$,*

$$\kappa(Q_{2k_1}, \dots, Q_{2k_r}) = O \left(n^{1 - \sum_{i=1}^r (k_i - 1)} \right). \quad (53)$$

Proof. Represent each Gaussian monomial by a vertex with $2k_i$ slots. Wick pairings give multigraphs on these vertices, allowing self-loops. Grouping pairings by their connected components is exactly the moment-cumulant recursion, so the cumulant is the sum of connected pairings.

There are $M = \sum_i k_i$ paired edges. In each covariance from (28), choose the row-equality term, the column-equality term, or the negative product term, denoted respectively R, C, and X. Let t be the number of X edges, and let ρ_R, ρ_C be the ranks of the R and C edge graphs on the r vertices; a self-loop has rank zero. Kronecker equalities leave $b = 2r - \rho_R - \rho_C$ free row and column index blocks. The total exponent of all normalized weights v_i is $2M + 2t$. A free block of exponent $d \geq 0$ has absolute sum at most $O(n^{1-d/2})$, since $\max_i |v_i| = O(n^{-1/2})$; this includes an unweighted block with $d = 0$. The colored term therefore is

$$O(n^{2r - \rho_R - \rho_C - M - t}).$$

Replace the R and C subgraphs by spanning forests in each of their components, and retain the X edges. Connectivity of the full graph is preserved, whence $\rho_R + \rho_C + t \geq r - 1$. The exponent is at most $r + 1 - M$, which is the one in (53). Only finitely many pairings and color choices occur at fixed degrees. This argument bounds their absolute sums, without using a favorable cancellation or a sign assumption on covariance entries. $\square$

10.2 Two finite contractions

Write $x = s_2$, $y = s_3$, $z = s_4$, and use

$$F = Q_6, \quad H = Q_8, \quad V = \mathbb{E}D^2, \quad K = \text{Cov}(Q_4, Q_6), \quad T = \kappa(Q_4, Q_4, Q_4) = \mathbb{E}D^3.$$

These local letters are unrelated to the phase characteristic function or the margin subspace used earlier. The bivariate Wick formula gives

$$K = \sum_{e,f} \left(h_e h_f^2 P_{ef}^2 + \frac{2}{3} h_f P_{ef}^4 \right). \quad (54)$$

For T , write $D = \sum_e (\frac{1}{2} h_e : Y_e^2 : + \frac{1}{12} : Y_e^4 :)$. The four possible triples of chaos degrees give

$$\begin{aligned} T = & \sum_{e,f,g} h_e h_f h_g P_{ef} P_{eg} P_{fg} + \frac{3}{2} \sum_{e,f,g} h_e h_f P_{eg}^2 P_{fg}^2 \\ & + 2 \sum_{e,f,g} h_e P_{ef} P_{eg} P_{fg}^3 + \sum_{e,f,g} P_{ef}^2 P_{eg}^2 P_{fg}^2. \end{aligned} \quad (55)$$

All cell indices are unrestricted. For example, the chaos degree triples $(2, 2, 2)$, $(2, 2, 4)$, $(2, 4, 4)$, and $(4, 4, 4)$ have respectively 8, 24, 192, and 1728 pairings for one ordering; the coefficients in (55) include the three possible positions in the two mixed cases and the coefficients in D .

For reproducibility, the complete polynomial reductions of these contractions are as follows:

$$\begin{aligned} K = & \frac{10}{3} nxy + \frac{34}{3} x^2 + 4y - \frac{52}{3} x^3 - 2x^4 + \frac{16}{3} x^3 y + \frac{148}{3} x^2 y - \frac{5}{3} x^2 y^2 \\ & - \frac{40}{3} x^2 z - \frac{40}{3} xy - \frac{76}{3} xy^2 + \frac{28}{3} xyz - \frac{64}{3} xz - 22y^2 + 40yz + \frac{16}{3} z - 10z^2, \end{aligned} \quad (56)$$

$$\begin{aligned} T = & 11nx^3 + 37x^2 + 2y - 104x^3 + 124x^4 + 40x^5 + \frac{11}{2} x^6 - 50x^4 y - 232x^3 y \\ & + 70x^3 z - 114x^2 y + 94x^2 y^2 + 50x^2 z + 46xy + 258xy^2 - 210xyz \\ & + 8xz - 22y^2 - 102yz + 4z + 90z^2. \end{aligned} \quad (57)$$

We specify an exact finite derivation so these are not identities inferred from interpolation. For a degree vector $(d_1, \dots, d_r)$, enumerate loop counts ℓ_i and nonnegative off-diagonal edge multiplicities e_{ij} with $2\ell_i + \sum_{j \neq i} e_{ij} = d_i$, retaining connected graphs. The number of slot pairings for each graph is

$$\frac{\prod_i d_i!}{\prod_i (2^{\ell_i} \ell_i!) \prod_{i < j} e_{ij}!}. \quad (58)$$

This follows by assigning the slots at each vertex to its incident edges, pairing its $2\ell_i$ loop slots, and matching slots across each pair of vertices. For every edge multiplicity e , choose $a + b + c = e$ R, C, X colors, with coefficient $(-1)^c e! / (a! b! c!)$. Merge row blocks along R edges and column blocks along C edges. At each endpoint an R edge adds one column weight power, a C edge one row power, and an X edge one of each; a loop adds twice at its endpoint. After merging, a block of even degree $2j$ contributes s_j and a block of odd degree contributes zero, by reflection. Here $s_0 = n$. Multiply by (58) and the appropriate q_k 's, and sum. This finite distributive expansion of (28) proves (56)–(57) for every $n \geq 2$. The package implements exactly this symbolic computation and checks the two polynomial coefficient lists independently.

As a shorter check of the terms actually needed asymptotically, there are two connected graphs for $(4, 6)$ and eight for $(4, 4, 4)$. Their contributions are shown below; each last-column entry is per labeled graph and omits only $O(n^{-3})$ terms, including all q_k factors.

Degree pattern and edges	Graphs	Pairings	Contribution per graph
(4, 6), two joining edges, loops 1, 2	1	540	$2nxy + 4y + 10x^2$
(4, 6), four joining edges, loops 0, 1	1	360	$\frac{4}{3}nxy + \frac{4}{3}x^2$
(4, 4, 4), double-edge star, leaf loops	3	864	$nx^3 + 7x^2$
(4, 4, 4), triangle, one loop each	1	1728	$2nx^3 + 2y + 12x^2$
(4, 4, 4), edges 1, 1, 3, one loop	3	1152	$\frac{4}{3}nx^3 + \frac{4}{3}x^2$
(4, 4, 4), double triangle, no loops	1	1728	$2nx^3$

It follows, also directly from the exact polynomials, that

$$K = \frac{10}{3}nxy + 4y + \frac{34}{3}x^2 + O(n^{-3}) = \frac{13176}{175n^2} + O(n^{-3}), \quad (59)$$

$$T = 11nx^3 + 2y + 37x^2 + O(n^{-3}) = \frac{167778}{875n^2} + O(n^{-3}). \quad (60)$$

10.3 A finite exponential expansion with an order three remainder

Equation (36) with $K = 4$ gives

$$\frac{C_n}{2n^2 G_n} = \mathbb{E}e^{-Q_4 - F - H} + O(n^{-3}).$$

Set $Z = D + F + H$. Its fixed moments obey $\|Z\|_p = O_p(n^{-1/2})$ and $Z \geq -m$. Taylor expansion of e^{-Z} through degree five therefore has expected remainder $O(\mathbb{E}|Z|^6) = O(n^{-3})$. To control every monomial in that finite polynomial more sharply than by its absolute moments, use the cumulant partitions of $\mathbb{E}D^a F^b H^c$. Singleton D blocks vanish. Every surviving partition has at most $\lfloor a/2 \rfloor + b + c$ blocks: blocks containing no F, H require at least two D 's, and each other block can be charged to one F or H . By Lemma 10.1, its product is

$$O(n^{-a-2b-3c+\#\text{blocks}}) = O(n^{-\lceil a/2 \rceil - b - 2c}).$$

In the second equality the right side is an upper bound; a partition can have fewer than the maximal number of blocks. Centering changes no cumulant of order at least two. In particular,

$$\mathbb{E}D^4 = 3V^2 + O(n^{-3}), \quad \mathbb{E}D^2 F = V\mathbb{E}F + O(n^{-3}), \quad \mathbb{E}F^2 = (\mathbb{E}F)^2 + O(n^{-3}).$$

Keeping the terms of orders at most two gives, with $\mu_F = \mathbb{E}F$ and $\mu_H = \mathbb{E}H$,

$$\begin{aligned} \mathbb{E}e^{-Q_4 - F - H} &= e^{-m} \left[1 - \mu_F - \mu_H + \frac{V}{2} + K + \frac{\mu_F^2}{2} - \frac{T}{6} \right. \\ &\quad \left. - \frac{V\mu_F}{2} + \frac{V^2}{8} + O(n^{-3}) \right]. \end{aligned} \quad (61)$$

All other degree-at-most-five terms are $O(n^{-3})$ by the displayed partition bound. No infinite cumulant series has been interchanged with an expectation.

10.4 Rational assembly

Centered power summation gives

$$x = \frac{9}{5n} - \frac{12}{5n^3} + O(n^{-5}), \quad y = \frac{27}{7n^2} + O(n^{-4}), \quad z = \frac{9}{n^3} + O(n^{-5}).$$

More generally, $s_r = n^{1-r}(3^r/(2r+1) + O(n^{-2}))$ for fixed r ; this follows by expanding centered even power sums, or by the midpoint sum for the fixed polynomial of degree $2r$. The mean of Q_8 is $\mu_H = (17/24) \sum_e h_e^4$, using the Gaussian eighth moment 105. Equations (33), (48), (50), (46), and the displayed fourth leverage sum give the coefficients

Quantity	Coefficient of n^{-1}	Coefficient of n^{-2}
$m - 7/5$	$-9/5$	$-39/100$
V	$243/25$	$-2691/175$
μ_F	$216/35$	$-2484/175$
μ_H	0	$8466/175$
K	0	$13176/175$
T	0	$167778/875$

Taking the ordinary logarithm of the finite expansion (61) cancels its square terms. Its first coefficient is $L_1 = 171/350$ as before, and the second is

$$L_2 = \frac{39}{100} - \frac{2691}{350} + \frac{2484}{175} - \frac{8466}{175} + \frac{13176}{175} - \frac{167778}{5250} = \frac{6483}{3500}.$$

Exponentiating this finite expansion gives

$$c_2 = L_2 + \frac{L_1^2}{2} = \frac{6483}{3500} + \frac{29241}{245000} = \frac{483051}{245000},$$

proving (52). The symbolic contraction computation and the independent raw-Gaussian-moment checks in the package use rational arithmetic; they do not use finite matrix counts to estimate this coefficient.

11 Parity aware inversion and integer thresholds

11.1 The logarithmic phase

Use $d = 12$ on odd indices and $d = 3$ on even indices, so that $S_n = n(n^2 - 1)/d$. Define

$$A_0 = \log 2, \quad B_0 = \log(2/\pi), \quad D_d = \log(2d/\pi), \quad E_d = -\log d - B_0/2 - 7/5. \quad (62)$$

Taking logarithms in Theorem 1.1 gives, on the specified parity,

$$\log C_n = A_0 n^2 - 3n \log n + D_d n + 3 \log n + E_d + \sum_{j=1}^J r_j n^{-j} + O_J(n^{-J-1}). \quad (63)$$

The r_j are rational and independent of parity. They are the formal coefficients of

$$-(n-1) \log(1 - n^{-2}) + \log \left(1 + \sum_{j \geq 1} c_j n^{-j} \right),$$

so in particular $r_1 = 1 + c_1 = 521/350$ and $r_2 = 6483/3500 - 1 = 2983/3500$. Each requested finite order involves only finitely many coefficients. Positivity needed for the logarithm follows from the leading equivalent.

11.2 Why the first crossing is eventually odd

The leading equivalent, evaluated at adjacent parities, gives

$$\frac{C_{2m}}{C_{2m-1}} \sim 2^{4m-1} \frac{2}{\pi} \frac{1}{S_{2m}} \left(\frac{S_{2m-1}}{S_{2m}} \right)^{2m-2}.$$

Here

$$S_{2m} = \frac{2m(2m-1)(2m+1)}{3}, \quad \frac{S_{2m-1}}{S_{2m}} = \frac{m-1}{2(2m+1)}.$$

Canceling the powers of four leaves (6), since $((2m-2)/(2m+1))^{2m-2} \rightarrow e^{-3}$. On either fixed parity,

$$\frac{C_{n+2}}{C_n} \sim 2^{4n+4} e^{-6} \left(\frac{2d}{\pi} \right)^2 n^{-6} \rightarrow \infty. \quad (64)$$

Thus each parity subsequence eventually increases strictly and tends to infinity, while every sufficiently large even term is below its preceding odd term. Once Y exceeds the maximum of the finitely many exceptional early terms, its first crossing cannot be even. This proves Corollary 1.3, without asserting a numerical onset.

11.3 The inverse of a smooth asymptotic phase

Let $L = \log Y$ and $X = \sqrt{L/A_0}$. A large real root of a smooth truncation of (63), for either d , has the expansion

$$\begin{aligned} x &= X + u + \frac{A_0 u^2 + 3u - 3 \log X - E_d}{2A_0 X} + O\left(\frac{\log^3 X}{X^2}\right), \\ u &= \frac{3 \log X - D_d}{2A_0}. \end{aligned} \quad (65)$$

To verify it, substitute $x = X + u + v/X$. The order- X terms cancel when $2A_0 u = 3 \log X - D_d$. The remaining constant terms are $2A_0 v - A_0 u^2 - 3u + 3 \log X + E_d$, which gives the displayed v . All omitted phase terms are $O(\log^3 X/X)$; the phase derivative is asymptotic to $2A_0 X$, giving the stated inverse error. Higher inverse coefficients follow recursively by formal substitution, using the finite-order remainders to justify any fixed truncation. The r_1/x term first affects a later inverse order than the X^{-1} term displayed here.

11.4 Rigorous odd integer envelopes

For the odd branch $d = 12$, let $M_J(x)$ be the right side of (63) truncated after $r_J x^{-J}$, omitting the sum if $J = 0$. There are constants K_J, n_J such that, for all odd $n \geq n_J$,

$$M_J(n) - K_J n^{-J-1} \leq \log C_n \leq M_J(n) + K_J n^{-J-1}. \quad (66)$$

Both bounding functions are strictly increasing for all sufficiently large real x , since their derivatives are asymptotic to $2A_0 x$. Let $x_+(L)$ and $x_-(L)$ be their large roots:

$$M_J(x_+) + K_J x_+^{-J-1} = L, \quad M_J(x_-) - K_J x_-^{-J-1} = L.$$

Then $x_+ \leq x_-$. Define

$$\text{ceil}_{\text{odd}}(x) = 2 \left\lceil \frac{x-1}{2} \right\rceil + 1,$$

the least odd integer not below x . For all sufficiently large Y ,

$$\text{ceil}_{\text{odd}}(x_+(\log Y)) \leq N(Y) \leq \text{ceil}_{\text{odd}}(x_-(\log Y)). \quad (67)$$

Indeed every large odd $n < x_+$ has its upper phase bound below L , while the odd integer $\text{ceil}_{\text{odd}}(x_-)$ has its lower phase bound at least L . The finite initial segment is excluded by increasing Y if necessary. This proof also covers an exact root at an odd integer, because the bounds use $\geq$ at a threshold.

The mean value theorem gives

$$x_-(L) - x_+(L) = O_J(X^{-J-2}).$$

Away from an odd integer within this shrinking interval, the two ceilings agree and determine $N(Y)$ exactly. When they straddle an odd integer, the asymptotic information intentionally leaves a rounding ambiguity. No uniform single rounded smooth formula follows at all thresholds.

An interpolation qualification. Formula (65) concerns a smooth asymptotic phase. Even on one parity branch, ordinary piecewise-linear interpolation across intervals of length two has a chord error of order one, since the smooth phase's second derivative tends to $2 \log 2$. This can produce an inverse error of order X^{-1} , precisely the scale of the displayed second correction. Interpolation between adjacent indices of different parities has the additional branch oscillation. Neither interpolation is silently assigned the smooth formula's X^{-1} coefficient. The envelope construction is the rigorous discrete inverse statement.

Remark 11.1 ([write] The inversions against the transseries volume, 6 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`. Its symbols are marked “vol” where they would collide with this report's ($h, E, t, f, \Lambda, \kappa, d$). Fix a parity, hence d , and put $\ell_X := \log X$.

- (a) *The smooth inverse (65) is a formal instance of `p0:thm:core-reversion`, after a change of variable, and of no Lambert theorem of the volume.* The core $\Phi_0(X) = A_0 X^2$ is admissible in the sense of `p0:def:core` ($\Phi'_0 = 2A_0 X > 0$ increases to $+\infty$ and $X\Phi'_0 \rightarrow \infty$), with core solution $X = \sqrt{L/A_0}$, the X of (65). It is neither the linear-logarithmic core $aX + b \log X$ nor the factorial core $X(\kappa_{\text{vol}} \log X + d_{\text{vol}})$, so (65) is not an instance of `p0:thm:lambert-core` or `p0:prop:factorial-core`, nor of `p0:thm:lambert-centered` or `p0:thm:flattening`, which concern the model $ax + b \log x + Q(1/x)$. Write the truncated phase of (63) as $M(x) = A_0 x^2 - 3x \log x + D_d x + 3 \log x + E_d + \sum_{j=1}^J r_j x^{-j}$ and substitute $x = X(1 + E_{\text{vol}})$, $t_{\text{vol}} = 1/X$, so that $\log x = \ell_X + \log(1 + E_{\text{vol}})$. Dividing $M(x) - L$ by X^2 gives exactly

$$\Lambda_{\text{vol}} E_{\text{vol}} + h_{\text{vol}}(E_{\text{vol}}) + f_{\text{vol}}(t_{\text{vol}}, E_{\text{vol}}) = 0, \quad \Lambda_{\text{vol}} = 2A_0, \quad h_{\text{vol}}(u) = A_0 u^2,$$

$$\begin{aligned} f_{\text{vol}}(t, u) = & t[D_d(1+u) - 3(1+u)(\ell_X + \log(1+u))] \\ & + t^2[3\ell_X + 3\log(1+u) + E_d] + \sum_{j=1}^J r_j t^{j+2}(1+u)^{-j}, \end{aligned}$$

over the commutative $\mathbb{Q}$ -algebra $R = \mathbb{R}[\ell_X]$, with ℓ_X an indeterminate and $\Lambda_{\text{vol}} = 2 \log 2$ invertible; $h_{\text{vol}} \in u^2 R[[u]]$ and $f_{\text{vol}} \in tR[[t, u]]$, as the theorem requires. Its unique solution $E_{\text{vol}} = \sum_{n \geq 1} e_n t^n$ has, by the triangular recurrence (`p0:eq:core-triangular`), $2A_0 e_1 = 3\ell_X - D_d$ and $2A_0 e_2 = A_0 e_1^2 + 3e_1 - 3\ell_X - E_d$ (at order t^2 the terms $-3e_1 \ell_X + D_d e_1$ equal

$-2A_0e_1^2$). So $e_1 = u$, $e_2 = v$, and $x = X + e_1 + e_2X^{-1} + \dots$ is (65) through its X^{-1} term. The source's "higher inverse coefficients follow recursively by formal substitution" is that recurrence; r_1 enters first at order t^3 , that is in the X^{-2} term of x , as the source says. The instance is formal only: `p0:thm:core-reversion` is a statement about formal series, and the analytic statement, that the large root of the truncated phase differs from the truncation through e_2 by $O(\log^3 X/X^2)$, is the source's own argument; for higher truncations see Question 8.

- (b) *The first threshold and the odd envelopes are analogues, not instances, of `p0:thm:staircase`.* The staircase of `p0:def:three-inverses` presupposes a sequence strictly increasing from some index on, and (C_n) is not: by (6), $C_{2m} < C_{2m-1}$ for all large m . Each parity subsequence is eventually strictly increasing by (64), and the proof of `p0:thm:staircase(4)` uses only such classwise monotonicity; read that way, (4) with $r = 2$ gives $N(Y) = \min(2\lceil \nu_0(Y)/2 \rceil, \text{ceil}_{\text{odd}}(\nu_1(Y)))$ for large Y , for any admissible interpolations of the two classes (the parity pattern of `p0:rem:parity-instances`), and Corollary 1.3 says that the odd candidate is the minimum. But (4) is stated under the standing hypothesis that (C_n) violates, so this is an extension of the volume's statement, not an instance. The envelope (67) is likewise an analogue of `p0:thm:staircase(2)`: $x_{\pm}$ are roots of two smooth envelopes evaluated at odd integers, not approximations, with a proved error bound, to the inverse of an interpolation (the paragraph above explains why an interpolated inverse may differ from the smooth one at order X^{-1}); the ambiguity when $\text{ceil}_{\text{odd}}(x_+) \neq \text{ceil}_{\text{odd}}(x_-)$ is the failure mode of (2).
- (c) *No continuous formula.* For all large k , $N(Y) = 2k + 1$ on $(C_{2k-1}, C_{2k+1}]$: $N(Y)$ is odd there by Corollary 1.3, the odd subsequence is eventually strictly increasing, and C_{2k-1} exceeds the finitely many early terms. So N jumps by 2 at $Y_0 = C_{2k-1}$, and for every continuous g , as in the proof of `p0:thm:staircase(5)`, $2 = |N(Y_0 + \varepsilon) - N(Y_0)| \leq |N(Y_0 + \varepsilon) - g(Y_0 + \varepsilon)| + |g(Y_0 + \varepsilon) - g(Y_0)| + |g(Y_0) - N(Y_0)|$; letting $\varepsilon \downarrow 0$ gives $\sup_{Y \geq Y_1} |N(Y) - g(Y)| \geq 1$ for every Y_1 . This makes precise the source's sentence that no uniform single rounded smooth formula follows.

No novelty is claimed for these inversions.

12 Finite counts and exact reproducible checks

12.1 An exact meet in the middle formula

Let

$$V_n = \{v \in \{0, 1\}^n : w^T v = 0\}, \quad b_n = |V_n|.$$

Every admissible matrix is a sequence of rows in V_n . Split the row weights into positive, negative, and, if n is odd, one zero weight. Reflection pairs the positive and negative lists. For a choice of rows $v_k \in V_n$ indexed by the positive weights k , form $s = \sum_{k>0} kv_k \in \mathbb{Z}^n$ and let $f_n(s)$ be the number of such choices producing s . Then

$$C_n = b_n^{n \bmod 2} \sum_{s \in \mathbb{Z}^n} f_n(s)^2. \quad (68)$$

The column constraints say that the positive and reflected-negative weighted row sums agree; their choices are independent and use the same list of weights. When present, the row of weight zero is any element of V_n . This proves (68) exactly.

The finite values are

n	b_n	C_n	Positive row tuples
1	2	2	1
2	2	2	2
3	4	16	4
4	4	16	16
5	8	512	64
6	8	512	512
7	20	323600	8000
8	18	144930	104976
9	52	1808243216	7311616

They agree with the displayed square array in OEIS A222959 [1]. The independent finite enumeration underlying the last line uses 7,311,616 positive weighted row tuples. The standard build’s bounded check range and any optional larger enumeration are documented explicitly in the code package; recorded $n = 9$ evidence is not mislabeled as a fresh default replay.

[write] **The $n = 9$ evidence and the program that is not shipped, 6 October 2026.** The package’s `DATA_SOURCES.md` and `data-README.md` (delivered `data/README.md`) say that “all nine diagonal values were independently checked by a C++ positive/negative-row meet-in-the-middle implementation”, and `data/reference.json` records $n = 9$ as such a “recorded independent computation”. That C++ program is not part of the delivered package, so this report cannot ship it, and the claim cannot be checked as stated. It is not needed: the shipped Python (`code/check_exact.py`, option `-max-n9`) recomputes $C_9 = 1808243216$ from the 7,311,616 tuples in about ten seconds on a copy of the delivered layout (rerun at the dossier and at the write), the intake’s own programs reproduce $C_1, \dots, C_9$, and the OEIS records the same values (Remark 12.1).

Remark 12.1 ([write] The OEIS records, and C_{10} , 6 October 2026). The entry A222959 [1] (revision #8, read on 6 October 2026) is named

T(n,k)=Number of nXk 0..1 arrays with every row and column least squares fitting to a zero slope straight line, with a single point array taken as having zero slope.

(quoted verbatim). Write $T_{959}(n, k)$ for this array, so that $C_n = T_{959}(n, n)$, and $C_1 = 2$ is the entry’s single-point convention. Its data lines list the array by antidiagonals, $T_{959}(n, k)$ at index $(n+k-1)(n+k-2)/2+n$ (the array is symmetric), and reach the diagonal only through $T_{959}(6, 6) = 512$; the value 8000 that occurs there is $T_{959}(5, 7)$, not a diagonal term. The comment “Table starts” displays the array for $n \leq 12$ rows, with the diagonal 2, 2, 16, 16, 512, 512, 323600, 144930, 1808243216 for $n \leq 9$, the values of the table above. The entry links Hardin’s b-file of 199 terms [12], which the source could not retrieve (HTTP 403) and the write could: its diagonal entries for $n \leq 9$ are the same, and its index 181 gives

$$C_{10} = T_{959}(10, 10) = 615667968.$$

A222955 [2] is the first column, $b_n = T_{959}(n, 1)$ (its cross-reference reads “Column 1 of A222959”); its data and b-file agree with the alphabet sizes of Section 12 for $n \leq 14$, and its formula line says only “An asymptotic formula for $a(n)$ can be found in Prodinger’s paper.”

Independent counts. A meet-in-the-middle program written for the write (not part of this report), which evaluates (68) with the sums s partitioned exactly by their first two coordinates, gives C_n for $2 \leq n \leq 10$; for $n \leq 9$ these are the table’s values, and for $n = 10$ ($b_{10} = 48$, $48^5 = 254,803,968$ positive row tuples) it gives 615667968, the b-file value. *One more ratio.* At $n = 10$ the ratio of C_n to $2^{n^2} G_n e^{-7/5}$ is 6670.0 (and 6241.9 after division by $1 + c_1/n + c_2/n^2$; the corresponding values at

$n = 7, 8, 9$ are 19.05, 3259.4 and 21.94). The even-index ratios grow from $n = 8$ to $n = 10$ while the odd ones stay near 22: at these sizes the counts are further from the asymptotic regime on even indices, which is consistent with the source’s warning and validates nothing. Nothing was submitted to the OEIS.

12.2 Why these values do not validate the asymptotic numerically

For $n = 7, 8, 9$, the ratios of the exact counts to $2^{n^2} G_n e^{-7/5}$ are approximately 21.15, 3558.9, and 23.66. These large discrepancies are strongly preasymptotic, not confirmation of numerical asymptotic accuracy. A simple admissible sector already explains some small-dimension structure: matrices palindromic in both rows and columns have $\lceil n/2 \rceil^2$ free binary entries, and hence

$$C_n \geq 2^{\lceil n/2 \rceil^2}. \quad (69)$$

Opposite weights cancel pairwise in each row and column, proving this bound. The sector is exponentially negligible relative to the asymptotic main count as $n \rightarrow \infty$, since the difference in leading binary exponents is $3n^2/4 + O(n)$, dominating $O(n \log n)$.

Small row alphabets also have accidental rank deficiencies: through $n = 14$, their sizes are

$$2, 2, 4, 4, 8, 8, 20, 18, 52, 48, 152, 138, 472, 428,$$

and their real linear ranks are

$$1, 1, 2, 2, 3, 3, 5, 5, 8, 9, 10, 11, 12, 13.$$

In particular fixed-width patterns from the first few cases cannot replace a growing-dimensional proof. These are exact finite observations, not hypotheses used by the analytic argument.

12.3 What the code establishes

The accompanying source archive contains the manuscript, source ledger, standard-library exact checks, recorded data with provenance, and a strict reproducible build and verification workflow. The mathematical checks use integer and rational arithmetic and explicit exceptions rather than Python assertions, so optimized Python does not disable them. They check finite projection identities, normalized moments, the first two coefficients, exact structural identities, and bounded direct enumeration. A symbolic connected Wick computation proves the two second-order contraction polynomials by exact coefficient collection; a separately implemented raw Gaussian moment recursion checks them for $n = 2, 3, 4, 5$. The builder and its negative tests check the declared file inventory, manifest integrity, output safety, and normal versus optimized replay. Exact ranges and commands are recorded in the package documentation and machine-readable outputs.

Finite checking is corroboration of finite algebra and implementation. It does not prove the global localization theorem, certify an eventual asymptotic threshold, evaluate the uncomputed c_j for $j \geq 3$, or constitute formal verification in a proof assistant. Reproducible PDF bytes are claimed only for a fixed installed TeX toolchain; independent toolchain versions may render the same source with different bytes.

[write] The package here, and the intake reruns, 6 October 2026. “The accompanying source archive” is the delivered one. In this report its programs are in `code/` (`matrix_exact.py`, `check_exact.py`, `derive_second_correction.py`, `verify_second_correction.py`, and the delivered root files `reproduce.py`, `build.py`, `verify_manifest.py`, `test_build.py`); the provenance record `code/PROVENANCE.json` is `data/PROVENANCE.json`; the generated receipts `generated/*.json` are `data/generated-*.json`; the code

and data READMEs are `code-README.md` and `data-README.md`; the manuscript is `article.tex`; the checksum manifest and the PDF are not shipped. The programs pin these files by path and SHA-256 and the manifest checker expects the closed delivered inventory, so they run only on a copy of the delivered layout (the report README gives the route). On a fresh extraction of the archive at the write (Windows, Python 3.14.4): `verify_manifest.py` passes (21 files); `reproduce.py` passes in about 12 s, and both its normal and optimized `exact_checks.json` equal the delivered receipt byte for byte; `test_build.py` passes with 321 tests (84 acceptance, 237 rejection), against 326 (242 rejection) in the delivered `generated/build_guards.json` (the five others, three named-pipe and two double-slash-path rejections, run only where `os.mkfifo` exists or `os.name` is `posix`); the optional `-max-n9` run reproduces C_9 . The PDF and archive builder was not run.

13 Primary source comparison and limits of the search

This section separates established predecessors from the specific theorem proved here. The comparison is based on the primary statements identified below, rather than titles alone. Its scope is a focused search completed on 4 October 2026; it is not exhaustive.

13.1 The sequence and one dimensional fair words

OEIS A222959 defines binary arrays with zero least-squares slope in each row and column, and displays the finite square counts used above [1]. The companion A222955 concerns the one-row condition [2]. The displayed array was available, and its square entries through nine were independently reproduced; the direct A222959 b-file was not available in the source retrieval. That access limitation is not evidence that the sequence lacks existing data.

Prodinger’s 1979 paper studies binary words with equal counts of scattered ab and ba , relates them to a centered weighted sign sum, and proves parity-sensitive one-dimensional asymptotics in Theorem 4, pp. 273–274 [3]. Richomme’s later primary treatment gives the weighted-sum characterization in Corollary 5.11, attributes the asymptotics to Prodinger in Theorem 5.12, and proves the equivalence with zero least-squares slope in Theorem 5.13 [4]. In the notation of the row alphabet above, the quoted one-dimensional leading forms are

$$b_{2m} \sim 2^{2m-1} \sqrt{\frac{3}{\pi}} m^{-3/2}, \quad b_{2m+1} \sim 2^{2m+1} \sqrt{\frac{3}{\pi}} m^{-3/2}.$$

The row interpretation and its asymptotics are therefore prior work. They do not by themselves count simultaneous row and column constraints in a square matrix, whose number of independent equations grows with n .

[write] **6 October 2026.** The A222959 b-file that the source could not retrieve is available and is used in Remark 12.1. The forms displayed above agree with Richomme’s Theorem 5.12, $b_n \sim 2^{2\lfloor(n-1)/2\rfloor+1} (3/\pi)^{1/2} \lfloor n/2 \rfloor^{-3/2}$, read by the write; Prodinger’s paper was not read.

Remark 13.1 ([write] The conjecture recorded in A222955 is a theorem, 6 October 2026). The entry A222955 [2] (revision #24, 17 March 2026, read on 6 October 2026) carries the comment

Conjecture: A binary word is counted iff it has the same sum of positions of 1’s as its reverse, or, equivalently, the same sum of partial sums as its reverse. -
Gus Wiseman, Jan 07 2023

(quoted verbatim). Richomme proved it [4]: his Theorem 5.13 identifies the zero-slope words with the fair words, and his Theorem 5.14 states that a binary word is fair exactly when the positions of one letter have the same sum in the word and in its mirror image; his introduction says the paper solves “a 2023 conjecture presented in” the entry. The entry still labels the comment a

conjecture. *A second, direct proof* of both forms, in the notation of Section 1: for $v \in \{0, 1\}^n$ put $|v| = \sum_j v_j$, $P(v) = \sum_j j v_j$ and $v_j^R = v_{n+1-j}$. Then $P(v^R) = \sum_j (n+1-j) v_j = (n+1)|v| - P(v)$, so $P(v) = P(v^R)$ if and only if $P(v) = (n+1)|v|/2$. For $n \geq 2$ the slope numerator of Section 1 is $\sum_j (j - (n+1)/2) v_j = P(v) - (n+1)|v|/2$, so the slope vanishes exactly when v and v^R have the same sum of positions of 1's. The sum of the partial sums of v is $\sum_{k=1}^n \sum_{i \leq k} v_i = \sum_i (n+1-i) v_i = P(v^R)$, and that of v^R is $P(v)$; so the second form is the same condition. For $n = 1$ both words are counted by the entry's convention and both equal their reverses, so the statement holds for every $n \geq 1$. In particular b_n counts the words with $w^T v = 0$, as Section 12 uses. Nothing was submitted to the OEIS.

13.2 Fair picture arrays have weaker aggregate conditions

Bera, Sriram, Nagar, Pan, and Subramanian study Parikh matrices of binary picture arrays [5]. Their Definition 5 of fairness requires equal totals of scattered ab and ba aggregated over all rows, and an analogous aggregate condition over columns. The surrounding results, including Theorem 4 and Sections 3–7, concern algebraic operations, concatenation, powers, and geometric properties. Requiring each individual row and column to be fair is stronger. For instance the 2×2 identity array has canceling aggregate row and column imbalances, but neither of its length-two rows is individually fair. This literature is relevant terminology and history, not a located asymptotic enumeration theorem for the present count.

13.3 Prescribed nullvectors and singular random matrices

Arratia and DeSalvo develop lower-bound expansions for singularity of Bernoulli sign matrices using nullvector templates associated with integer partitions [6]. Their definitions, Theorem 1, Proposition 9, Lemma 3, and the full or almost-full support discussion in Section 5 address prescribed nullvector events and their interactions. The fixed-template constants in Proposition 9 depend on the template support and one-row concentration. Propositions 10–11 condition a prescribed nullvector against exclusions such as sparse opposite-side templates; they do not count two prescribed growing arithmetic-progression nullvectors simultaneously. The present inference is limited: those estimates cannot be silently made uniform when both the support and the coefficients grow with n . They are conceptual predecessors, not an inspected matching count equivalent.

13.4 Maximum entropy Gaussian and Edgeworth frameworks

The binary counting theorem, Theorem 2.6, of Barvinok and Hartigan's Gaussian paper has quantitative covariance and boundedness hypotheses [7]. Its parameter λ is a lower bound for the quadratic Fourier form $q(t) = \frac{1}{2} t^T \text{Cov } t$. At fixed relative accuracy it must dominate a constant times $\theta^2 d^2 \log N$, where θ controls the constraint-column ℓ^1 norms. The natural saturated coordinates here fail this sufficient condition. Delete one column-margin coordinate whose weight is 1; then $d = 2n - 1$, $N = n^2$, and $\theta = \Omega(n)$. Remove that coordinate from the full gauge vector $(w, -w)$. The resulting vector has squared norm $2S - 1$, and its image under the reduced design transpose has squared norm S . The Bernoulli covariance is one fourth of the reduced design Gram matrix, so

$$\lambda_{\min}(\text{Cov}) \leq \frac{S}{4(2S-1)}, \quad \lambda_{\min}(q) \leq \frac{S}{8(2S-1)} = O(1).$$

Thus the required domination fails in these coordinates. The factor of two between the bounds is solely the source's quadratic-form normalization. No claim is made about every possible reformulation.

The paper also explains that these sufficient conditions do not establish ordinary two-way-table Gaussian asymptotics.

The later Edgeworth paper states a general program with explicit conditions I–V in Theorem 1, pp. 5–7 [8]. Condition V is precisely a relative negligibility requirement for the Fourier integral outside a small central region. Its ordinary contingency-margin specialization does not verify that condition for growing signed arithmetic-progression coefficients. We credit the maximum-entropy, Fourier, and Wick correction framework; we prove the necessary problem-specific global estimate in Sections 4–6 instead of treating the framework as an automatic theorem.

13.5 General high dimensional lattice counting

Kuperberg, Lovett, and Peled give a general local-limit and counting framework for regular combinatorial structures [9], including Theorems 2.4–2.5 and the basis-free Theorem 4.15. One hypothesis requires a coordinate permutation symmetry group acting transitively on the ground set. Here the ground set consists of the n^2 cells, and the constraint space is E . A coordinate permutation preserving E also preserves its orthogonal projection diagonal. But $h_{ij} = a_i + a_j - a_i a_j$ varies across cells for large n , ruling out that transitivity. Their counting theorem’s size lower bound of order $\dim(E)^6$, even before boundedness and logarithmic factors, also exceeds n^2 in this regime. These actual hypothesis obstructions explain why that important general precedent does not directly establish our theorem.

13.6 Other nearby matrix models

Harrison and Miller’s weighted binary matrix model assigns nonuniform cell odds to matrices with ordinary prescribed row and column sums [10]. Its weights are probabilities, not growing signed coefficients inside each margin equation. Definition (1) and Section 4.2 establish the different scope: importance sampling with ordinary margins and associated approximations.

Aw’s admixed arrays involve paired binary matrices with uniform unweighted row and paired-column sums [11]. The inspected model, Theorem 4.3 setup, reduced cosine-squared integral, and local expansion use coefficients one in the phase constraints. The local quartic approach is relevant, but no general growing-weight global-localization theorem is imported from that setting. In particular, a modular cosine bound is not extended to unwrapped real phases outside a central interval.

13.7 Scoped conclusion

The focused comparison did not locate a primary theorem matching the simultaneous centered-arithmetic-progression left and right nullvector count, the factor $e^{-7/5}$, or its rational all-fixed-orders expansion. This is a limited negative search result. The safe mathematical contribution stated here is a complete application and extension of established Fourier and higher-order Gaussian enumeration methods, including a fully explicit defect-core localization proof. Absolute priority remains open. The package’s source ledger records links, inspected theorem scope, and retrieval limits; no novelty assertion rests on a search-result title or absence from a catalog.

14 Limitations and further research

[write] **Added 6 October 2026, batch 108.** This section serves as the report’s “Further questions and research” section under Vladimir Reshetnikov’s standing rule of 4 October 2026: every claim of the source

that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. Items 1–7 are the source’s and state their own source, sketch and gap; its other non-claims are collected in Section 1.1. The intake and the write found no false claim in the source. Two further statements of the source are not proved in full: the sentence of Section 11 on higher inverse coefficients (now Question 8), and the package’s statement that a C++ program checked all nine counts (that program is not shipped; the counts are reproduced, note in Section 12). Item 5 is advanced by Proposition 14.1 below, which sharpens the palindromic bound (69), and item 1 by the count C_{10} (Remark 12.1): at $n = 10$ the ratio to the main term is about 6670, so the even branch is still far from its asymptotic regime.

The following questions are not resolved by the theorem or by its code.

1. **Practical error and onset.** The proof is eventual and deliberately uses tiny absolute thresholds. A useful explicit error bound at accessible n would require substantially sharper localization and local estimates. The small exact values cannot locate that onset.
2. **Higher explicit coefficients.** Formula (42) gives every fixed coefficient by a finite computation, and c_1, c_2 have been evaluated here. The supplied contraction-class computation treats the two contractions needed at second order. Extending that implementation systematically could make $c_3, c_4, \dots$ practical; those coefficients are not supplied as computed results.
3. **Optimal truncation.** Bounds uniform in a growing order J , convergence, Borel summability, and exponentially improved expansions all lie beyond this fixed-order argument.
4. **Other weight lists and rectangular limits.** The lattice and Gaussian core formulas are rectangular, but the present global theorem uses both dense arithmetic-progression circuits and square-scale defect costs. Extending it to rectangular aspect ratios or other deterministic nullvectors requires checking new arithmetic stability and localization hypotheses.
5. **Transition from exceptional finite structure.** The exact palindromic sector and row-alphabet rank defects suggest a useful study of intermediate dimensions. They explain caution, but do not supply a rigorous crossover law.
6. **Integer inverse boundaries.** The envelopes shrink to one odd integer except near threshold coincidences. Sharper asymptotics do not alone remove every possible rounding ambiguity; exact comparison or explicit remainder bounds are needed there.
7. **Literature and formalization.** A broader primary-source search could find additional predecessors. A formal proof assistant development would require encoding the global inverse lemmas, Haar normalization, and uniform analytic remainders; the current standard-library checks are not such a formalization.

For $n, k \geq 1$ let $T_{959}(n, k)$ be the number of $n \times k$ binary arrays whose n rows (of length k) and k columns (of length n) all have zero least-squares slope, length one counting as zero slope; so $T_{959}(n, n) = C_n$, $T_{959}(n, 1) = b_n$, and T_{959} is the array of A222959 (Remark 12.1).

Proposition 14.1 ([write] Palindromic rows, narrow arrays). *(1) For all $n, k \geq 1$, $T_{959}(n, k) \geq b_n^{\lceil k/2 \rceil}$.*

(2) Equality holds in (1) for every $n \geq 1$ when $k \leq 6$, and fails for every $n \geq 1$ when $k \geq 7$.

(3) $C_n \geq b_n^{\lceil n/2 \rceil} \geq 2^{\lceil n/2 \rceil^2}$ for $n \geq 1$, and $b_n^{\lceil n/2 \rceil} / (2^{n^2} G_n) \rightarrow 0$.

Proof. A palindrome $v \in \{0, 1\}^k$ has zero slope: for $k \geq 2$ the positions j and $k + 1 - j$ carry opposite weights $\pm(j - (k + 1)/2)$ and equal entries, so the slope numerator of Section 1 vanishes; for $k = 1$ by convention. There are $2^{\lceil k/2 \rceil}$ palindromes of length k , so $b_k \geq 2^{\lceil k/2 \rceil}$.

(1) An $n \times k$ array all of whose rows are palindromes has column j equal to column $k + 1 - j$. Such an array is counted exactly when each column lies in V_n (its rows have zero slope already), and its columns $1, \dots, \lceil k/2 \rceil$ can be chosen freely in V_n and determine it. This gives $b_n^{\lceil k/2 \rceil}$ distinct counted arrays.

(2) For $k \leq 6$, $b_k = 2, 2, 4, 4, 8, 8 = 2^{\lceil k/2 \rceil}$ (the table of Section 12, and A222955), so the zero-slope words of length k are exactly the palindromes; every counted array then has palindromic rows, and (1) is an equality. For $k \geq 7$ let u have its ones at positions $\{1, 4, k-2, k-1\}$, four distinct positions since $k-2 \geq 5$. Then $\sum_j j u_j = 2k+2 = (k+1)|u|/2$, so u has zero slope (by the computation in Remark 13.1), and u is not a palindrome: its mirror image has its ones at $\{2, 3, k-3, k\}$, which contains 2. The $n \times k$ array all of whose rows equal u is counted, because its columns are constant and a constant column has zero slope ($\sum_i w_i = 0$ for $n \geq 2$; convention for $n = 1$); its rows are not palindromes, so it is not among the arrays of (1), and the inequality is strict.

(3) Take $k = n$ in (1) and use $b_n \geq 2^{\lceil n/2 \rceil}$. Since $b_n \leq 2^n$, $b_n^{\lceil n/2 \rceil} \leq 2^{n(n+1)/2}$, while $\log(2^{n^2} G_n) = n^2 \log 2 - 3n \log n + O(n)$ by (3); the ratio is at most $\exp(-n^2 \log 2/2 + O(n \log n)) \rightarrow 0$. $\square$

Part (2) explains the equal columns of the displayed OEIS array ($T_{959}(n, 1) = T_{959}(n, 2) = b_n$, $T_{959}(n, 3) = T_{959}(n, 4) = b_n^2$, $T_{959}(n, 5) = T_{959}(n, 6) = b_n^3$; for instance $T_{959}(7, 5) = 20^3 = 8000$, the value noted in Remark 12.1). Part (3) improves the palindromic bound (69): at $n = 7, 8, 9, 10$ it gives 160000, 104976, 380204032 and 254803968, against 2^{16} , 2^{16} , 2^{25} , 2^{25} and the counts 323600, 144930, 1808243216, 615667968. Like that bound, it is negligible as $n \rightarrow \infty$; it says that the row-palindromic sector alone already carries 72% of C_8 and 41% of C_{10} , part of the small-dimension structure of item 5.

8. **Higher terms of the smooth inverse** (source: Section 11, “Higher inverse coefficients follow recursively by formal substitution, using the finite-order remainders to justify any fixed truncation”). The formal coefficients are those of the instance of `p0:thm:core-reversion` in Remark 11.1(a), polynomials in $\log X$. The source proves the error $O(\log^3 X/X^2)$ for the truncation through the X^{-1} term. Open, as written: for each fixed N , an error bound for the truncation through e_N for the large root of the truncated phase. Missing: the written estimate, which the source gives in one sentence (substitute the truncation into the phase, bound the defect, divide by the phase derivative $\sim 2A_0 X$).
9. **Fixed width** (added by the write, not a claim of the source). For fixed $k \geq 7$, the asymptotics of $T_{959}(n, k)$ as $n \rightarrow \infty$: a k -variable local limit over the finite row alphabet V_k , with the parity of n entering through the column weights. Proposition 14.1 settles $k \leq 6$ exactly; for $k \geq 7$ only the lower bound $b_n^{\lceil k/2 \rceil}$, strict, is known here. Missing: the local limit theorem for this lattice, as in Section 2 with a fixed number of columns.

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